Molecular representations for drug discovery

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Abstract

Boosted by the incredible advancement of machine learning algorithms, machine learning aided drug discovery has quickly become a central topic in the field over the past decade. In this chapter, rather than focusing on algorithms, molecular representations typically seen in the field are revisited. The categories of molecular representations consist of four types of data "modalities": sequential, topological, spatial, and temporal modalities, inspired by the categorization of primary, secondary, tertiary, and ensemble protein structures. Each modality is reviewed with examples from the literature. Moreover, knowledge graphs are discussed for representation learning, along with multimodal fusion techniques aimed at leveraging the strengths of each modality. With successful stories in other fields, it is foreseeable that by combining the right domain knowledge with the correct combination of modalities and the proper algorithm, the next breakthrough of machine learning aided drug discovery is on the horizon.

Bios

Leili Zhang is a staff research scientist at IBM Thomas J Watson Research Center since 2015. He leads the small molecule research challenge within IBM, which aims to build AI foundation models to accelerate small molecule drug discovery. His research interests further extend to disease mechanism simulations, protein folding mechanisms, protein-protein interactions, membrane proteins and free energy simulations. Leili received PhD from University of Wisconsin-Madison. He has served as guest editors for journals such as Frontiers in Molecular Biosciences and International Journal of Molecular sciences.

Alex Golts is a ML/DL researcher and engineer at the Multimodal AI for Healthcare group at IBM Research – Israel since 2021. Since 2022 he contributes to the building of large multi-modal biochemical foundation models and fine tunes them for a variety of predictive and generative downstream tasks related to drug discovery. During 2021-2022 he focused on learned models for medical imaging and clinical data. He is a core contributor to the FuseMedML and Fuse-Drug open source projects. His interest in Machine Learning began in 2014. Over the years he developed and applied learning based solutions to a variety of challenging problems in different domains, with an emphasis on Computer Vision. He received his M.Sc. in Electrical Engineering from the Technion in 2015.

Vanessa Lopez is a senior researcher scientist and manager at IBM Research Europe, since 2012. Her research explores the potential of enhancing multimodal AI foundation models with knowledge graphs to assists human experts unlock scientific discoveries in complex domains, such drug discovery. She and her team investigate extracting knowledge from heterogeneous sources and graph representation learning for effectively combining multimodal rich factual knowledge for predictive tasks. Before joining IBM, she was a researcher at KMI, Open University, where she pioneered research on semantic Question Answering systems. She was a trainee at the European Space Agency and holds a master's degree in computer engineering from the Technical University of Madrid. Over the years, she has served on organisational and programme committees, and has co-authored over 100 publications, with more than 5100 citations.

1 Introduction

For machine learning aided drug discovery tasks, molecules and targets are first converted to machine processable data formats before being processed with models such as linear regression, decision tree, random forest, support vector machine, autoencoders, convolutional neural network, recurrent neural network, generative adversarial network, transformers, etc. These machine processable data are referred to as molecular representations. Categorization of data representations vary through literature. For example, categories based on human reading habits include text, graph, image, and video.[1] Categories based on biological and biomedical concepts include DNA, RNA, protein, small molecules, disease textual descriptions, biomedical networks, etc.[2] In this chapter, definitions in Consonni et al.[3] are updated to better categorize common molecular representations based on physical understanding. Typical molecular representations in the field are focally presented using this updated definition. Machine learning algorithms are purposefully de-emphasized, because they have been extensively covered by other reviews[4-6], and drug discovery tasks, discussed in additional reviews[7-9].

It is first necessary to define the categories of molecular representations used in this chapter. The term "modality" was originally used in drug discovery as "chemical modalities" (or "therapeutical modalities"), which refer to ways of treatment (with small molecules, biologics, physical therapy, etc.). With the heavy influence from computer sciences (specifically, multimodal fusion techniques[10]), data modality is introduced to the field with a different annotation. A data modality is defined as a group of representations that is detected with the same sensor and follow the same data formatting rules and therefore, no translation is needed when the machine is reading a molecular representation within the same modality. Subsequently, categories of descriptors introduced by Consonni et al.[3] can be reintroduced in terms of data modalities. It is notable that Consonni et al.'s original definitions of 1D, 2D, 3D, and 4D descriptors were loosely defined with given examples.

Here, these definitions are reformulated using notions from protein structures (**Figure 1**). Specifically, primary structure (sequence) corresponds with sequential (1D) modality; secondary structure corresponds with topological (2D) modality; tertiary structure corresponds with spatial (3D) modality; and conformational ensemble corresponds with temporal (4D) modality. To determine if a modality is 1D, the data needs to be derived from information of neighboring atom pairs that are chemically bonded. Therefore, protein sequences and molecular SMILES naturally fall into this category. To determine if a modality is 2D, the criteria is whether it explicitly contains or derives from extended bonded information such as angle and dihedral angles. To determine if a modality is 3D, the criteria is whether it explicitly contains or derives from spatial distances or contacts. To determine if a modality is 4D, the criteria is whether it explicitly contains or derives from an ensemble of conformations.

With these criteria, a sequential (1D) modality now represents molecules using atom names and atom properties, with implied connectivity between adjacent atoms. Naturally, representations such as protein sequences, SMILES, SMARTS, etc. fall in the sequential modality. A topological (2D) modality represents molecules with atoms and their local environment. Therefore, representations such as circular fingerprints (ECFP fingerprints etc.), 2D images, and 2D molecular graphs fall in this category. A spatial (3D) modality takes advantage of coordinate information of the molecules (therefore, it is conformation-sensitive and symmetry-sensitive). As a result, distance/contact matrices, 3D fingerprints, 3D molecular graphs, and 3D images fall in this category. A temporal (4D) modality adds time-dependent information to the representation, including contacts in a time series, conformations in a time series, conformational flexibility, and entropy terms. Finally, Knowledge Graphs are added as structured representations of domain knowledge that capture entities (e.g., proteins) along with their textual attributes and relationships (edges) connecting these entities. One major difference between the updated definition and Consonni et al.[3] is that Consonni et al. categorizes SMILES as a 2D representation and MACCS fingerprints as a 1D representation, while the updated definition categorizes SMILES as a 1D representation and MACCS fingerprints as a 2D representation. We argue that MACCS fingerprints categorize fragments in molecules similar to secondary structure categorizations of residues, while SMILES can be seen as a molecule sequence. Therefore, the updated definition better conforms with the current existing categorization of protein structural properties.

In the following sections, each modality of molecular representations is reviewed as defined above. Representative studies will be discussed especially for representations that have gained high popularity.

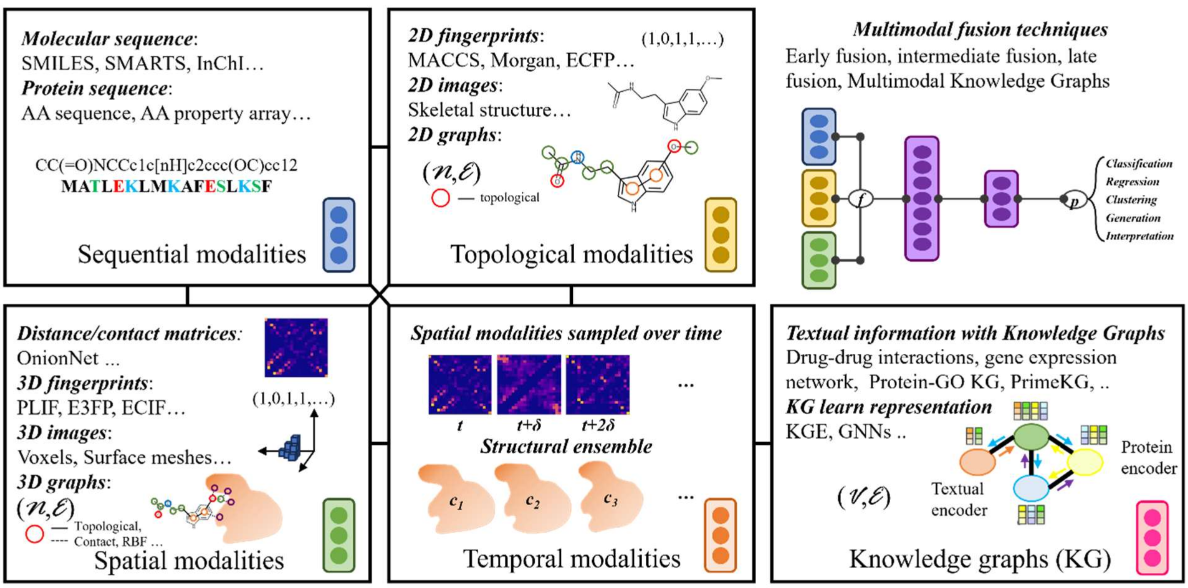

Figure 1. Data modalities (see main text for definitions) are exemplified and illustrated with typical molecular representations in their categories. Note that, (N (or V), E) stands for the nodes (or vertices) and edges defining a graph. RBF stands for radial basis functions. Symbol t stands for initial time of sampling

and symbol δ stands for the time interval between samples. Symbols c_1, c_2, and c_3 stand for conformations sampled in a structural ensemble. In "Multimodal fusion techniques", f stands for multimodality aggregation functions and p stands for learning functions.

2 Sequential modalities

A molecule is a 3D collection of atoms held together by chemical bonds. As such, although it is natural to represent molecules with information from higher dimensionalities, adopting these data can be challenging due to data availability, high memory and time complexity. For this reason, many AI applications opt for simpler, more compact, linear sequence representations such as different types of strings that may attempt to encode the connection pattern of the molecular graph. The field of natural language processing (NLP) has enjoyed great advances in recent years, notably through the development of large language models (LLMs)[11-13]. Using string representations for representing molecules, in addition to being efficient, allows to leverage these important advances, applying them in the field of drug discovery[14-15].

2.1 String representations for small molecules

The most popular string notation to represent small molecules is Simplified Molecular Input Line Entry System (SMILES)[16]. It is non-unique (the same compound can be represented by more than one SMILES string) and unambiguous (a given SMILES string can correspond to a single compound). A SMILES string is obtained by assigning a number to each atom in the molecule and traversing the molecular graph using that order. Different atom ordering can lead to a different SMILES string (non-uniqueness). Hence, it makes sense to enumerate different SMILES for data augmentation[17]. Inset in **Figure 1** (under "Topological modalities") shows a skeletal structure plot of a Melatonin molecule, whose possible SMILES representation is CC(=O)NCCc1c[nH]c2ccc(OC)cc12. MolFormer[18] developed by Ross et al. is a transformer-based language model which is trained on over 1 billion SMILES from ZINC and PubChem dataset. It is a foundation model (a machine learning model that's pretrained on a large set of data) that can be finetuned on smaller set of data for prediction tasks such as spectra predictions, solubility predictions and toxicity predictions.

In an attempt to make SMILES unique, several canonical versions were proposed[19-20], as well as other types of modifications and extensions like SMARTS (SMILES Arbitrary Target Specification)[21] and SELFIES (Self-referencing embedded strings)[22] addressing different challenges. SMARTS does not address the non-uniqueness of SMILES but offers an extension, with additional symbols to help specifying substructure patterns. SELFIES on the other hand, focuses on providing a robust representation that will always represent a valid molecule. One reason that SMILES is non-unique is that the order in which atoms and bonds are specified can be different. Canonical implementations of SMILES follow a set of rules to ensure the same ordering is used every time, ensuring uniqueness. One such implementation is available in the RDKit[23] library. An alternative representation of Melatonin traversing the molecular graph in a different order is CC(=O)NCCC1=CNc2c1cc(OC)cc2. Applying RDKit's Chem package MolFromSmiles function followed by MolToSmiles we obtain the canonical representation COc1ccc2[nH]cc(CCNC(C)=O)c2c1, for both alternatives.

Given the non-uniqueness of SMILES, as well as the inherent redundancy in such large datasets it is important to meaningfully split the data to avoid overfitting of machine learning models. To this end, a range of split strategies are often employed, ranging from simply making sure the exact same compound is not found in both train and test folds, to more robust splits taking into account molecule similarity.

MoleculeNet[24] implements a scaffold splitting method, splitting the data into folds based on their 2D structural frameworks.

Another popular representation is the open source International Chemistry Identifier (InChI)[25], and their unique hashed version, the InChIKey, used for web and library searching[26]. Unlike SMILES, InChIs are unique but can be ambiguous. In many drug discovery applications SMILES or InChI based small molecule representations are combined with representations for macromolecules, such as proteins (discussed in the next section). One such important use case is developing learning based models for the prediction of binding affinity, the likelihood of a small molecule drug candidate to successfully bind to a protein target of interest. For this purpose, datasets are curated that contain pairs of possibly binding molecules. One such large curated dataset in which small molecules representations contain both SMILES and InChI is available in Golts et al.[27] An open source implementation applying a Protein Language Model (PLM) based Drug Target Interaction (DTI) model[28] on this dataset, has been made available[29]. It is based on fuse-drug, a molecular biochemistry library for drug discovery and repurposing, which is a component of FuseMedML, a more general Python framework for accelerating ML based discovery in the medical field[30]. Both SMILES and InChIs are atom-based. As such, they encode the exact structure of a molecule.

2.2 String representations for proteins

While representations for small molecules typically focus on the atomic level, macro-molecules such as proteins are commonly defined using their nucleotide and amino acid (AA) sequences. AAs, made up of an amine group, a carboxyl group, and a side chain, are the building blocks of peptides and proteins. They are commonly represented by a one-letter symbol, or a three-letter abbreviation. The known genetic code consists of only 22 proteinogenic AAs (20 common AAs and 2 rare AAs) commonly represented by one-letter symbols.

Clustering and partitioning of protein sequences have been proven to be powerful tools when parsing protein sequences because of the homology among proteins that share a common evolutionary origin. To avoid data leakage and overfitting, maximizing (clustering method) or minimizing (partitioning method) the similarity between entities in the training dataset and the hold-out evaluation set is crucial. Multiple sequences alignment (MSA) is a category of alignment/clustering method that evaluates the molecular phylogeny of an unknown sequence and estimates the evolutionary similarity and divergence between sequences[31]. The foundation of MSA-based methods relies on the notion that subsequences that carry key structures and functions tend to be conserved during evolution[32]. The usefulness of MSA has been proven by numerous studies with the highlight of AlphaFold[33] which takes MSA as the pre-requisite for its structure prediction tasks. Hestia[34] is a unified framework for creating similarity-based partitions for different biochemical sequential data types, including protein sequences, DNA sequences, and small molecule string representations (SMILES), and for meaningful split of the data into training and evaluation subsets. The effect of partitioning strategy and threshold depend both on the specific prediction task and the biochemical data type, being more sensitive for tasks for which homology is important, like enzymatic activity classification.

Protein sequences have been trained with transformers in studies like ESM (Evolutionary scale modeling)[35] and ProteinBERT[36] for function predictions. With masked reconstructions, they learn the likelihood that a particular amino acid appears in a sequence, capturing co-evolutionary and inter-residue contact information from the raw sequences alone.[35, 37]

DTI tasks commonly focus on protein targets. In dataset curated by Golts et al.,[27] and subsequently the implementation[30], small molecules are paired with such targets, represented by sequences of AAs. Such datasets lend themselves for models that use both small molecules and macro-molecules representations as input. In Born et al.[38] a similar task is solved with the protein sequence only restricted to a key region called the "active site" that can be defined in multiple ways.

In addition to the straightforward representation of proteins as sequences of AA symbol letters, there are representations which combine atomic and sequence information as well as allow inferring one from the other[39-41]. Among these representations, HPNN[42] adds a vector to each residue that represents its attention to all other residues.

A thorough review written by David et al.[43] discusses in detail especially about other well-known string notations.

[31-33343] 3 Topological modalities

While sequential modalities mainly focus on the bonds of immediate neighbors and typically adopt a language-like model, topological modalities take advantage of extended bond information or take the form of a molecular image. Infrastructures of vector-based machine learning models and image-based machine learning models are typically paired with this modality. For the simplicity of the chapter structure, another typical 2D modality – ***2D molecular graph*** is combined with ***3D molecular graph*** in the next section.

3.1 2D Fingerprints

Unlike atom-based string representations, representations can also include different physical properties as descriptors of a molecule[3], which can be categorized as molecular fingerprints. There are many types of fingerprints. Broadly, they can be classified according to dimensionality[44]. 0D fingerprints encode global molecular properties (e.g. molecular weight), 1D fingerprints utilize the bond information of immediate neighbors within a molecule, 2D fingerprints include extended connectivity information within the molecule, and 3D fingerprints are derived from 3D coordinates of the atoms in the molecule.

In addition to dimensionality, fingerprints can also be further classified into two types: structural keys and hashed fingerprints. Structural keys are binary strings encoding the absence or presence of different chemical groups. A popular example of 2D fingerprints with structural keys are Molecular ACCess System (MACCS) keys[45], also called MDL keys, named after the company that developed them: MDL Information Systems. A subset of 166 MACCS keys is available via popular open-source cheminformatics software libraries, such as RDKit[23]. Each such key encodes as a binary fingerprint a specific structural feature or arrangement of atoms in a molecule.

Hashed fingerprints are hashed vectors encoding physicochemical or structural properties mapped from the molecular graph. Hashed fingerprints may be classified into topological/path-based fingerprints (such as Daylight fingerprint[46]) and circular fingerprints (discussed below), according to the way by which the molecular fragments are enumerated.

Circular fingerprints consider the 2D circular environment of each atom up to a given diameter. Examples of circular fingerprints are extended-connectivity fingerprints (ECFPs)[47] and Morgan fingerprints. By selecting different values for the maximum diameter of the circular atom neighborhood, different flavors of ECFPs may be generated. Most commonly, a value of 4 or 6 is used for the diameter, generating the

ECFP4 and ECFP6 fingerprints, respectively. A variant of ECFPs that encodes the functionality or roles of atoms, are functional-class fingerprints (FCFPs)[47].

3.2 2D Images

Image representations for molecules are used primarily for visualization purposes, while some research works have utilized them as the form of input for AI models. This is motivated primarily by the impressive success of deep neural networks demonstrated across computer vision applications[48-51].

As a 2D image, a molecule is often represented by its skeletal structure (see **Figure 1** – Topological modalities). A standardization of the layout and rendering properties of molecule images is challenging, both for the purpose of visualization and AI based computation.

Skeletal drawings of molecules were shown to enable, via a deep Convolutional Neural Network (CNN), prediction of chemical properties, on par with models based on expert developed features, as shown in Chemception[52]. A similar approach was used in Meyer et al.[53] for classification of molecules into drug function classes. Here, the processing of molecule images by a CNN was also combined with molecular fingerprints input to a random forest classifier. In DEEPScreen[54], a similar approach was proposed for DTI. Images of drug candidate molecules were fed into a CNN to predict binary activity with a given protein target. ImageMol[55] is a foundation model pretrained on 10 million skeletal drawings with CNN. Subsequently, this foundation model is fine-tuned on SARS-CoV-2 dataset for DTI predictions. Post-hoc attention is calculated on molecule images using Grad-CAM[56] for explainability.

4 Spatial modalities

In physical simulations, enthalpy terms are calculated with spatial coordinates and force field. Naturally, representations that take advantage of spatial information have been considered as input for machine learning algorithms. For tasks that predict structures, spatial representations are used as labels and to calculate metrics that evaluate the model predictions. Based on our survey, there are roughly four categories which can be counted as spatial representations: distance/contact matrices, 3D fingerprints, 3D molecular graphs, and 3D images. Drug discovery workflows utilizing spatial modalities are typically referred to as Structure-based drug discovery (SBDD). Additionally, tasks such as protein structure predictions, small molecule conformation predictions, and crystal structure predictions naturally require spatial modalities with the aim to predict spatial information. Some representative studies are discussed using these representations in the following paragraphs.

4.1 Distance/contact matrices

It is natural to construct a coordinate matrix from known structures to take advantage of 3D information. However, scalar property predictions (such as affinity predictions, solubility predictions, toxicity predictions, synthesizability predictions, protein pocket identification, etc.) require the input data to be rotationally and translationally invariant (i.e., to satisfy SE(3) symmetry), which raw 3D coordinates of structures do not satisfy. While it is possible to make neural networks SE(3)-invariant[57], the more common strategy is to pre-process the input data to be rotationally and translationally invariant or to augment the dataset with manual translations and rotations. One way to pre-process the 3D coordinates to satisfy SE(3) symmetry is to convert coordinates to distances, which results in distance matrices. Authors' experience with using distance matrices as features along with various neural networks is that continuous distances often perform worse than binned distances. This observation is exemplified by the predominance of binned distance matrices seen in literature, which will be discussed below. A simple binning method is a step function, where a cutoff value is set at *c*, and the contact is counted if the distance is smaller than *c*. Based upon this, smoothened functions or stacked contact matrices using multiple cutoff values have also been proposed.

A notable example in this category is AlphaFold[33], an algorithm that predicts protein structures. To featurize the protein 3D structures, AlphaFold uses "distogram", a set of matrices of binned distance distribution between beta carbon atoms with 39 cutoff values. Additional features include multiple sequence alignment (MSA), amino acid type, residue index, backbone and side chain torsion angles and several control variables. Notably, a novel neural network "Evoformer" was constructed in this study to exchange information between MSA and distograms.

HIP-NN[58] is another example that utilizes distance matrix with a hard sphere cutoff = 15 Bohr (~7.9 Å). HIP-NN is constructed with the inspiration from many-body expansion where hierarchical energy terms with learned parameters are calculated with the distance matrix

RF-Score[59] is a protein-ligand binding affinity predictor that uses "coarse-grained" protein-ligand contacts as features. RF-Score featurizes 3D protein-ligand complexes with a set of atom-atom contact numbers with a single cutoff value at 12 Å. The coarse-grained characteristics stem from the way RF-Score groups atoms, where both proteins and ligands are coarse-grained to 9 common atom types: C, N, O, F, P, S, Cl, Br, I. All contacts within 12 Å are tallied for each pair, resulting a maximum of 9x9 = 81 dimensional features. In practice, features with insufficient data are deleted to save storage space.

OnionNet[60] is another example that uses coarse-grained contact matrices to predict protein-ligand affinity. In this study, 60 cutoffs are chosen between 8 atom types: C, N, O, H, P, S, HAX, DU, where HAX includes F, Cl, Br, and I and DU represents all other atoms.

4.2 3D fingerprints

The distinction between 3D fingerprints and 2D fingerprints is obvious: 3D fingerprints utilize the structure information that 2D fingerprints often omit, taking into account the spatial arrangement of atoms and how they are positioned relatively to each other in the 3D space. 3D fingerprints often incorporate the coarse-grained contact numbers similar to the quantities used in RF-Score. In some ways, 3D fingerprints can be seen as an expansion of the contact number representations. However, it is debatable if all 3D distance matrices can be converted to 2D contact fingerprints without information loss, especially for fine-grained distance matrices used in applications like AlphaFold. In this section, a few representative 3D fingerprints in this active research field are selected for discussion.

NNScore[61] uses a 3D fingerprint with 194 dimensional features. These features include hydrogen bond contacts (cutoff value is set at 2 Å) and other close contacts (cutoff value is set at 4 Å). Additionally, electrostatic interaction energy for all contacts within 4 Å, the number of atoms in an atom type, and the number of rotatable bonds in the ligand are also included. Comparing to the 9 types of atoms RF-Score uses, NNScore incorporated a more fine-grained atom type system from AutoDock Vina[62] which identifies hydrogen bond donors and acceptors (For example, HD is a hydrogen-bond-donor-type hydrogen atom and SA is a hydrogen-bond-acceptor-type sulfur atom).

The atom-atom contact in Extended connectivity interaction features (ECIF)[63] is more fine-grained than NNScore, which has 22 different atom types from proteins and 70 different atom types from ligands. The contact number feature therefore has 1,540 dimensions. Additionally, ECIF incorporates 170 molecular descriptors from RDKit[64].

Protein-ligand interaction fingerprint (PLIF) from MOE suite[65] contains information such as side chain hydrogen bonds, backbone hydrogen bonds, solvent hydrogen bonds, ionic interactions, metal binding interactions and arene interactions[66]. Similar approaches have also been developed such as SPLIF[67] which expands protein-ligand atom pairs with 2D circular fingerprints based on PLIF, PLIP[68] which provides 3D protein-ligand fingerprints from a web service or command line, and ProLIF[69] is a python interface that interfaces well with RDKit[64] and MDAnalysis[70].

Unlike the protein-ligand complex 3D fingerprints described above, ligand-only 3D fingerprints have also been developed, such as E3FP[71]. E3FP draws inspiration from ECFP but instead of circles representing atom connectivity, E3FP adds stereochemical identifiers by dividing spheres around an atom to octants.

QSAR-inspired 3D fingerprints have also been proposed. Khani et al.[72] uses principal component analysis (PCA) on the well-established 3D-QSAR model CoMFA (the comparative molecular field analysis) to extract ligand features and predicts IC50 value with support vector regression model.

Another subcategory of 3D fingerprints is inspired by molecular dynamics and molecular modeling. Ji et al.[73] constructed 3D fingerprints based on the decomposed energy terms calculated using MM-GBSA or explicit water molecules (TIP3P). Although energy minimization simulations and molecular dynamics simulations are performed to obtain the snapshot for energy decomposition, there is no time-dependent features in the fingerprints.

A representative 3D fingerprint inspired by algebraic topology is TopologyNet[74], which uses topological fingerprints (betti-0, betti-1, betti-2 numbers) of both protein and ligand heavy atoms and their interactions (hence categorized as 3D in the updated definition). Element-specific topological fingerprints are calculated, where heavy atoms are categorized to 4 and 9 for proteins and ligands, respectively. DG-GL[75] calculates "interactive curvatures" by constructing Riemannian manifolds for different atom type pairs. DC-GBT[76] converts the bipartite graphs (protein-ligand interactive matrices) to Dowker complexes with Riemann Zeta functions.

4.3 3D images

Although not easily comprehensive to human, 3D images can be seen as an extended 2D images to computers. Note that 3D images are not rotational invariant and therefore do not satisfy the SE(3) symmetry. In practice, data is often augmented by rotations of the images as initial input.

Ragoza et al.[77] converts protein and ligand complexes into voxels with 24x24x24 Å^3 grid and a resolution of 0.5 Å. Atoms are plotted with RGB-like channels based on their smina[78] atom types, with 16 protein atom types and 18 ligand atom types. Occupancy of an atom in a voxel is calculated with a combined Gaussian-quadratic density function based on their locality and van der Waals radius. The resulting 3D images are used to train CNN to predict protein-ligand interactions. Random rotations and translations are adopted to augment the data to overcome overfitting.

Similarly, DeepSite[79] divides the space into voxels and uses 8 channels of 16x16x16 voxels to represent different atom types of protein-ligand complex in space. However, instead of the smina atom types, these 8 channels of DeepSite correspond to chemical properties including hydrophobic, aromatic, hydrogen bond acceptor, hydrogen bond donor, positive ionizable, negative ionizable, metal atoms, respectively, with an additional channel of excluded volume for all atoms. The occupancy of an atom in a voxel is calculated using an exponential equation with their van der Waals radii. Subsequently, a CNN is trained with these 3D images as input to predict protein binding sites. This representation is reused in KDEEP[80] to predict protein-ligand binding affinity and Ligdream[81] for molecule generation.

Pafnucy[82] augments the voxel information with 19 features describing an atom, representing atom types, hybridization, heavy valence, hetero valence, hydrophobicity, aromaticity, hydrogen bond acceptor, hydrogen bond donor, ring atom, partial charge and molecular type (ligand or protein). This multichannel 3D image is paired with a CNN to predict protein-ligand binding affinity.

Instead of focusing on the bulk properties, MaSIF[83] calculates the surface properties of 3D protein structures, where predetermined fingerprints are calculated for all vertices on the surface mesh. These fingerprints include shape index, distance-dependent curvature, hydropathy index, continuum electrostatics, the location of free electrons and proton donors, radial coordinates, and angular coordinates. The vertex is labeled as interfacial if it is not accessible by solvent. The resulting surface mesh representation is paired with an image-based CNN to predict protein-ligand binding sites and protein-protein binding sites.

Similar to MaSIF, PINet[84] constructs surface properties of 3D protein structures where vertices contain information of Cartesian coordinates, electrostatics and hydrophobicity. The vertex is labeled as interfacial if it is within 2 Å of the surface of its binding partner. The resulting surface mesh representation is paired with an image-based CNN to predict protein-protein binding sites.

4.4 2D and 3D Molecular Graphs

Molecular graphs, alongside the original Graph Neural Network (GNN)[85], were created for structures such as molecules, images and a subset of the web. Molecular graphs store information in nodes and edges (N, E), where nodes (N) store information about the representing moieties (atoms or residues), and edges (E) store information about the connectivity (neighboring moieties, bond types and bond properties, etc.). It is noteworthy that graphs have flexible amount of feature dimensions, especially in the edges, where the amount of information varies based on how many neighbors a moiety has. The information of each node is propagated before being aggregated for tasks such as graph classification, graph clustering, graph matching, node classification, etc.[86] Graph Attention Network (GAT)[87] and Molecular Attention Transformer (MAT)[88] has also been developed to further strengthen the interpretability of GNN tasks. Note that the discussed graphs here include both 2D and 3D molecular graphs for the simplicity of writing structure. The difference between a 2D and 3D molecular graph is whether 3D coordinate information has been used to construct the graph (either in nodes or edges).

Undirected graphs are predominant in current molecular graph applications. For example, AquaSol[89] uses a minimalistic undirected graph representation which only includes atom types and bond types of the ligands (therefore a 2D molecular graph). They show that molecular solubility can be predicted well by combining undirected graph representation with recursive neural networks (RNNs). This study establishes that undirected graph representations are sufficient for molecule information representation and only challenged recently by attempts of using directed graph representations[90-91].

Weave[92] proposed a more expanded undirected graph of ligands with 27 node descriptors, including atom type, chirality, formal charge, partial charge, ring size, hybridization, hydrogen bonding and aromaticity, and 12 edge descriptors, including bond type, graph distance (not Euclidean distance), and whether two atoms are in the same ring (therefore a 2D molecular graph). Paired with graph convolution network (GCN)[93], Weave graphs are shown to work well with affinity and toxicity prediction tasks.

Cartesian coordinates are often used to augment 2D molecular graph representations, resulting in 3D molecular graphs. For example, DeepChem[94] expands upon Weave graphs by adding RDKit descriptors and 3D coordinate information. Based upon ligand-only DeepChem graphs, a 3D protein-ligand interaction graph that includes interacting residues within cutoff = 4 Å has also been proposed to rank docked poses[95]. DGraphDTA[96] constructed an undirected 2D ligand graph and a 3D protein graph to predict protein-ligand binding affinity. The 3D protein graph used in DGraphDTA has residue properties stored in nodes, and contact map information (with cutoff = 8 Å for optimized performance) stored in edges. L-Net[97] is a generative model for ligand designs that uses an undirected 3D ligand-only graph networks. L-Net graph contains atom type and Cartesian coordinate information in nodes, bond type information in edges. Due to the SE(3) non-equivariant nature of Cartesian coordinates, L-Net incorporates a "local coordinate system" that alleviates the issues raised from tasks of molecular generation. A protein folding algorithm, RoseTTAFold[98], utilizes 3D protein graphs where nodes represent residues in proteins and edges featurize various spatial relationships between residues. The information was communicated with graph transformer-based architecture between MSA and residue graphs. Some 3D molecular graphs focus on surface properties. For example, HoloProt[99] designed an undirected 2D ligand graph and a 3D surface-aware protein graph to predict protein-ligand binding affinity and protein functions. The 3D surface-aware protein graph has residue name, secondary structure, binary solvent accessible label and binary residue hydrophobicity stored in nodes, and residue-residue distances, Ramachandran angles stored in edges.

SchNet[100]-related method is another subcategory within molecular graphs, often referred to as geometric neural networks. In short, SchNet is an undirected 3D ligand representation that contains atom property and Cartesian coordinates information. Cartesian coordinates are then converted to distances and radial distribution functions in the form of radial basis functions to achieve SE(3) symmetry. Based upon SchNet, G-SchNet[101] directly utilizes pairwise distances between atoms to achieve both conformation-sensitive HOMO-LUMO (highest occupied molecular orbital–lowest unoccupied molecular orbital) energy gap predictions and molecules generation. 3D-Graph-SBDD[102] adds context-aware information for ligands (e.g., types of protein atoms surround ligand atoms) to generate molecules in known protein pockets. DimeNet[103] proposes to use (1) directed message passing based on angles between triplets of atoms (therefore, a directed graph) and (2) Fourier-Bessel basis functions and dramatically reduces the edge dimensions required for radial basis functions in SchNet. SpinConv[104] extends the inclusion of triplets of atoms by using spin convolutional filter on spatially neighboring atoms. PhysNet[105] adds an attention mechanism based on a physics-informed distance decay function. SphereNet[106] proposes spherical message passing scheme and included bond length, bond angles and dihedral angles in the Fourier-Bessel basis functions.

In the above paragraphs, a few publications are listed with notable graph innovations in the thriving research field of graph representations in drug discovery. Other reviews have focused on specific tasks such as generative models[107] and predictive models[108].

5 Temporal modalities

Time in molecular dynamics (MD) simulation plays a crucial role in estimating the entropy of a conformation and ligand binding kinetics. Machine learning methods have been applied to analyze molecular dynamics (MD) simulation in a number of studies[109]. MD simulation has also been used as a validation method (for tasks such as stability validation and free energy calculation) alongside machine learning methods[110-111]. In contrast, it is much less common to use featurized MD trajectory data (i.e., properties calculated with MD trajectory analysis tools) with machine learning for drug discovery tasks (affinity prediction, pose prediction, etc.). With the addition of the time dimension, it is of interest to see (1) whether it is of importance for drug discovery tasks, and (2) how to extract relevant temporal information from trajectory data.

Time-dependent fingerprints are proposed by MD-IFP[112] which utilizes PLIF and two dynamic features: (1) number of water molecules in the first solvation shell around the ligand, and (2) root mean square displacement (RMSD) from the reference (bound) position of the ligand. These 4D fingerprints are calculated for hundreds of snapshots collected from tens of short MD simulations (1 ps) before a k-means clustering is applied. The subsequent clusters are used to calculate persistent time, which has a good correlation with experimental results.

Time-dependent contact matrices are used as temporal modalities in some applications. For example, Ribeiro et al.[113] used time-dependent contact features (order parameters, implemented from RAVE[114]) with a variational autoencoder architecture[115] to select a linear combination of crucial contact features for protein-ligand interactions. These features are then validated by meta dynamics to observe infrequent protein-ligand dissociations that normally has timescale of minutes or longer. In this study, 20 ns of production runs (in MD simulations) are performed for data collection.

Another example of time-dependent contact matrices is CASTELO[116], where "temporal contact matrices" are constructed: half of the matrix is time-dependent ligand-residue contact number matrix and the other half is contact number delta matrix from t to $t+\delta$. Paired with unsupervised convolutional variational autoencoder (CVAE) and HDBSCAN clustering method, the authors identify hot spots within molecules that are deemed unstable for protein-ligand binding, and therefore, ideal moieties to change to improve binding affinity. In this study, data are collected from 50-100 ns of production runs. The same approach has been found to be as effective for protein-antibody interactions[117].

Time-dependent graphs are proposed by MD-Graph[118] that is based on DeepChem graphs[95] calculated for MD trajectories, where each frame is represented by a concatenated graph with a 2D ligand-only graph and a 3D protein-ligand interaction graph with 12 cutoff distances. The predicted softmax values are averaged over all frames to calculate the loss function. Data in this study are collected from 70-310 ns of production runs. The resulting MD-Graphs are then trained with GCN to predict HLA-peptide complex immunogenicity (without the presence of T cell receptors).

6 Knowledge Graphs

Accumulated scientific knowledge forms the foundation upon which informed decision making is built, particularly in the life science and therapeutics domains, scientific data is spread across multiple sources and its largely heterogeneous, encompassing multi-omics data, publicly available knowledge bases, experimental, pharmacological measurements, clinical data, and scientific literature. Therefore, there is growing interest in developing methods that can integrate such diverse data from multiple sources and modalities into ML and AI models. However, that requires learning an effective representation of the underlying data and its inherent biological linkages and complexities.

In the field of scientific discovery for therapeutics, the existent pre-trained sequence (1D) or structured based (2D-3D) models frequently overlook incorporating the wealth of biological structural knowledge available in curated and typically interconnected databases, such as Uniprot[119], Drugbank[120], ChEMBL[121], BindingDB[122], PharmaGKB[123], among others, to discover new knowledge. The challenge lies in effectively harnessing all this available factual knowledge to enhance representation learning and existent foundation models.

In recent approaches, Knowledge Graphs (KGs) are emerging as a tool to support not only the integration of rich factual knowledge from heterogeneous sources of knowledge but also to facilitate multimodal learning[124]. By leveraging the graph topology (relationships), KGs can help aligning the different embedding spaces across entities with different modalities and heterogeneous sources.

Prior research in the field of protein representation learning indicates that augmenting protein representations with additional information from human-curated domain knowledge, in the form of functional and structural annotations, can yield improvements in downstream learning tasks. A first effort in this direction was OntoProtein[125], which injected comprehensive textual data from the gene ontology GO[126], with information on the functions of genes, into a pre-trained Protein Language Models (PLM) for sequences. OntoProtein fine-tuned the PLM by reconstructing masked amino acids while minimizing the embedding distance between the contextual representation of proteins and associated GO functional annotations. Their results demonstrated utility in classification tasks, such as predicting protein-protein interactions, protein function, and contact prediction; but underperformed in regression tasks like homology, fluorescence, and stability. The gains, albeit present, are relatively modest compared to pretrained models using very large-scale corpora, since the knowledge can only cover a small subset. KeAP[127] is a more granular token-level approach than OntoProtein, where non-masked amino acids iteratively query the associated knowledge tokens to gather information (from GO) for the restoration of masked amino acids though cross-attention. Unlike OntoProtein, which employs both contrastive learning and masked modelling, KeAP training process is guided solely by the mask token reconstruction objective. ProtST[128] employs a dataset of protein sequences augmented with textual property descriptions from biomedical texts, it jointly trains a PLM alongside a biomedical language model. Protein function prediction, defined as the task of predicting a given protein with multiple and potentially hierarchical classes, as functions defined in GO, has also seen recent attention[129-130]. GOLabeler[131] focuses on extracting features, ensemble learning and automatic feature learning of proteins using various kinds of protein sequence information. DeepGraphGO[132] constructs a network of proteins and learns protein features through a Graph Neural Network. DeepGOZero[133] benefits from exploiting inter-function (class) relationships defined in GO, it does so by incorporating formal semantics, like class hierarchy, disjointness axioms and complex class restrictions, as additional constraints for training a multi-label classifier for protein function prediction.

These initial findings on PLMs enriched with textual descriptions using gene ontology annotations, exhibit promising enhancements in protein-related knowledge discovery tasks. However, they fall short of harnessing rich factual heterogeneous sources of data. Effectively learning a knowledge-enhanced representation often requires the incorporation of diverse knowledge sources, as information from a single source usually does not provide sufficient data[134-135].

Life sciences, encompassing chemistry, biology and medicine, generate and utilize vast amounts of scientific data, much of which is inherently relational or well-suited to be structured as graphs, with many interrelationships and dependencies. For this reason, and to facilitate knowledge integration, exchange and processing, graph-based innovations are often adopted within the life sciences and biomedical domain[136-138]. Knowledge Graphs (KGs) have long been used to offer an interpretable way to organize and combine background knowledge from diverse sources, making them powerful tools for representing intricate biomedical knowledge, including molecular interactions, signaling pathways, disease co-morbidities, and more, from heterogeneous data sources[138].

KGs organize structured knowledge using entities, relations, and attributes with well defined meanings (e.g, labels). A KG can be formally described as a directed labeled graph $G = (V, E)$, where vertices (V) or nodes with unique identifiers represent real-world entities of interest (e.g., proteins, genes, compounds, cellular components, pathways), while edges (E) represent relationships between the entities in V (e.g., binds, associates, type of, etc.) or datatype attributes of an entity (e.g, molecular function, the mass of a molecule, or the description of a protein). These are expressed as statements in the form of RDF triples such as (subject, predicate, object).

In graph representation learning, the graph's topology (entities and relations) is leveraged to create compact vector embeddings, making it possible to incorporate structured knowledge into machine learning applications. Through nonlinear transformations, high-dimensional information about a node's graph neighborhood is condensed into low-dimensional vectors, ensuring that nodes with similar characteristics are embedded close together within the vector space. A scoring function measures the plausibility of triples in KG, with higher scores for true triples and lower score for false/corrupted ones. These optimized representations are subsequently used to train downstream models for various tasks.

Traditionally KG embedding (KGE) models were often transductive[139] and not well-suited for inductive link prediction – essential in a drug discovery scenario involving the prediction for nodes not seen in the training). Inductive Graph Neural Network (GNN) methods are well-suited to capture cross-modal dependencies and diverse types of interactions between heterogeneous entity types through geometric relationships. They can learn a representation of entities (e.g., for which many edges may exists as both subjects and objects with other entities), edges or subgraphs in an inductive fashion. In link prediction, this entails predicting the existence of a link between two entities, even when one or both entities are unseen during training. This is achieved by reasoning about their connections to other known nodes or by reasoning about their attributes features, similar to those of nodes seen during training[140]. GNNs accomplish this by iteratively aggregating information from neighboring nodes (through a process called message passing) and employing scoring functions to optimize learned embeddings for downstream tasks.

GNNs have been applied to drug discovery[141], drug repurposing for COVID-19[142], or predicting polypharmacy side effects[143]. For instance, TxGNN[134] predicts drug indications/contraindications for rare diseases using a GNN pre-trained on PrimeKG[144]. PrimeKG is a large heterogeneous KG centred around diseases and describing relations between diseases, genes and therapeutic candidates, constructed from various knowledge bases. TxGNN generates a signature vector for each disease, based on its neighboring proteins, exposure and other biomedical entities. This allows for the computation of a disease similarity and the prediction of drug indication/contraindication, particularly for rare (poorly characterized) diseases.

7 Choice of data modality and multimodal fusion

Drug discovery entails exploring an extremely large space of potential candidates. AI can play a crucial role in accelerating drug discovery, predictive and generative models can be used to narrow down the most promising candidates (e.g., small molecules that can bind to a target protein) before embarking on expensive experimentation. The key to leveraging predictive and generative models for candidate solution generation lies in learning an effective representation of protein targets, molecules and diseases, among other entities. Representation learning employs techniques such as deep learning and nonlinear dimensionality reduction to create low-dimensional vector embeddings that encode essential information about diverse entities, such as proteins and molecules.

As seen in the previous sections, research focused on learning representations for protein and molecules has largely remained unimodal, based on large, self-supervised language models that are independently trained on vast databases containing millions of protein sequences or linear notations for small molecules. Models like ESM[35] and ProteinBERT[36] for protein sequences, or Molformer[18] for the molecule simplified molecular-input line-entry system (SMILES), are transformer-based Language Models (PLM) typically trained on masked reconstruction. These representations are then fine-tuned using limited labelled data for tasks such prediction protein contact[37], protein structure[145], function[37, 146], protein-protein interactions[147], or predicting the binding affinity between drugs and targets[15, 148-149]. Structured-based pretrained models go beyond 1D sequential tokens or 2D topology graphs, incorporating 3D structures to enable 3D geometry prediction or generation, as demonstrated by AlphaFold[33] for proteins and Uni-Mol[150] for small molecules.

With these successes, two questions arise: (1) “which data modality should one choose?” and (2) “can one combine multiple representations to improve the predictive/generative/… power by leveraging the strength and complementary signals from different modalities?”

There is no consensus in the field as in which data modality leads to generally more superior model performances as of now. One hypothesis is that a task (such as secondary structure prediction) may require a certain combination of molecular representation and machine learning algorithm for optimized performance. To find out which combination is optimal, it is crucial to provide impartial benchmarking processes to compare data modalities and machine learning algorithms (as suggested in our outlook in 8.9). One of the purposes of writing this chapter is to provide categories of data modalities differentiated by major physical differences (dimensionalities), so that systematic benchmarking can be attempted to compare data modalities and their combination. However, this is not to suggest that modalities with higher dimensionality are more superior than modalities with lower dimensionality. For example, molecular modelling inspired 3D fingerprints and algebraic topology inspired 3D fingerprints have not been implemented by temporal modalities. Time, as an extra dimensionality, has not been systematically proven to be helpful in predictive tasks. Moreover, even though 3D graphs can be seen as a combination of 2D fingerprints (stored in nodes) and 3D fingerprints (stored in edges), 3D graphs may not always outperform the constituent 2D/3D fingerprints (unpublished data and Jiang et al.[151]). This may be due to the “curse of dimensionality” as suggested by studies[152]: when the amount of data does not exceed the threshold needed for the data modality, overfitting or underfitting may occur. Practically, how to choose data modalities may solely depend on what and how much data is available. For example, if no structural data is available, while it is possible to rely on AlphaFold and small molecule structure prediction tools to generate structures, using only sequential or topological modalities might be a safer option.

While combining some data modalities may lead to data redundancy (for example, 3D graphs and their constituent 2D/3D fingerprints), others may provide independent/orthogonal information that may lead to better performing machine learning models. The algorithm that combines data modalities is a multimodal fusion algorithm. Some representative studies are reviewed in the following sections, focusing on non-KG modalities and KG modality.

7.1 Multimodality with non-KG methods

Foundation models (FMs) in biomedicine have predominantly centered around single data modalities, yet scientific data for drug discovery is inherently multi-modal. In biology, proteins targeting specific diseases and ligands can be described through all representations discussed above, including sequences, 2D/3D images, 2D/3D fingerprints, graphs, 3D coordinate information, textual attributes outlining their function, or categorical attributes, such as organisms and classifications etc. These modalities paired with machine learning algorithms are adopted to predict labels such as affinity, toxicity, as well as textual properties describing their function or indications, or generative new molecular entities. Information redundancy, caused by correlation between modalities, has been a major issue for multimodal machine learning algorithms. Stahlschmidt et al.[10] has laid out multimodal fusion strategies in three categories: early fusion, intermediate fusion, and late fusion. Early fusion strategies take as input a vector that simply concatenates each input data modality. Intermediate fusion strategies first learn a marginal representation of each modality to discover correlations within modalities and subsequently fuse these inside the latent space to learn joint representations. In this category, "homogeneous" intermediate fusion utilizes the same types of networks to generate marginal network representations and process learned joint representations, while "heterogeneous" intermediate fusion utilizes different types of networks. Finally, late fusion strategies combine decisions made by sub-models for each modality. In the following section, some algorithms not covered by Stahlschmidt et al. are discussed using these terms.

Early fusion models can be exemplified by Zhang et al.[153] that uses a multimodal deep belief network (DBN) to combine 1D sequence, 2D fingerprints (secondary structure in specific), 3D fingerprints (tertiary structure) to predict binding preferences of RNA-binding proteins. In this study, 1D sequences and 2D sequences are concatenated in shallow layers and 3D fingerprints are directly concatenated. Another example is GraphMVP[154], which combines 2D ligand graphs and 3D geometric graphs by using contrastive learning to reconstruct 3D graph from a 2D graph thus connecting the modalities. In this study, GraghMVP is shown to work to predict properties such as toxicity and binding affinity. In a follow-up study, GEOSSL-DDM[155], denoising score matching (DSM) is adopted to estimate distances between modalities, achieving better performance than GraphMVP.

An example of intermediate fusion study is MDeePred[156], which combines the latent space representations of 2D ligand fingerprints (ECFP4) and 1D protein sequence-based energy matrices in deep layers. It is shown that the fusion strategy performs slightly better than using only ligand fingerprints for binding affinity predictions. CPAC[157] combines 1D protein sequence-based representation (HPNN) and 3D protein graphs with intermediate fusion strategy to predict protein-drug binding affinity. In addition to latent space representation concatenation, the authors add cross-interaction introduced by Tan and Bansal[158].

A comparison of different fusion methods is conducted by Jones et al.[159]. In this study, 3D images and 3D graphs are combined with various fusion strategies, where 3D images are processed with 3D-CNN, 3D graphs are processed with SG-CNN (spatial graph neural network) and their latent space representations are combined in shallow layers (early fusion), deep layers (intermediate fusion) or at the decision level (late fusion). Among these, early fusion strategy is found to outperform the other approaches for the affinity prediction task, while late fusion strategy performs on average as well as the intermediate fusion strategy.

7.2 Multimodality for knowledge-enhanced representation learning

Research on Multimodal KG (MKG) is finding increasing application beyond computer vision and language modeling into biomedical networks[124, 138, 160]. In a MKG, entities may convey information about their modality, with typical examples including text, protein sequences, SMILES, images, 3D structures, numerical and categorical values. As any other KG, MKGs can encompass explicit labeled relations across multi-modal attributes for an entity, such as the description of a protein function or its average mass, and entities, such as interactions between proteins, complex pharmacological measurements, or other measurements like binding affinity between a protein and a given molecule, conditions or diseases associated to a molecule, etc. An effective learned representation must capture correspondences between the variety of data modalities in the knowledge for accurate predictions.

Some of the key challenges for representation learning using MKGs include effectively utilizing multimodal information and multimodal fusion, in particular from two or more modalities, to make predictions (e.g., classification, regression or link prediction), and the presence of missing modalities[124, 136, 161]. Inductive approaches are essential for modelling MKGs that encompass a variety of data modalities, since assuming that all entities have been observed during training is impractical. Learning a distinct vector for each entity and using enumeration for all possible attribute multimodal values to predict links is usually infeasible[140].

Recent inductive approaches that leverage a MKG to combine knowledge from diverse sources and modalities include OtterKnowledge[160] and BioBridge[162]. Both approaches use pretrained large single-modal models to compute initial embeddings for each modality and learn how to transform or fuse different modalities in an MKG, while keeping all the single-modal FMs fixed. In the case of OtterKnowledge, it also learns a representation for each entity in the graph from an arbitrary number of modalities known for that entity and its neighbouring entities in the graph. The KGs are just used for training. During inference, these knowledge-enhanced pre-trained models can be applied to machine learning downstream tasks to improve prediction accuracy.

The advantages behind OtterKnowledge and BioBridge in leveraging existent encoders, as opposed to creating a unified model for all modalities, are: (1) existent pretrained models are specialized in extracting distinct features that are relevant to that specific modality, leading to competitive and meaningful representations; (2) training a unified model requires cross-modals datasets of similar size, as heterogeneity, missing and imbalanced amounts of modalities may result in suboptimal learning; (3) training a single model will require a substantially larger training dataset to generalize to unseen entities. For example, ESM[35] is trained on close to 250M protein sequences from UniParc datasets, and Molformer[18] is trained on 1.1B unlabeled SMILES sequences from PubChem and ZINC datasets. In contrast, BindingDB has just approximately 20K binding affinity measures for 110 protein targets and around 11K small molecule ligands.

BioBridge[162] uses PrimeKG[144] to conduct cross-modal alignment of the embedding space for unimodal or single modality FMs during training. Specifically, it employs triples between entity types, such as proteins, molecules, biological processes, molecular functions, cellular components and diseases, categorized into three modalities: protein sequence, SMILES and natural language for all the other entity types. For each modality, BioBridge uses the models ESM2-3B[163], UniMol[127] and PubMedBERT[164] respectively. These triples, consisting of a head, and edge and a tail, are employed to learn a transformation across pairwise modalities in a triple. This is achieved by training a vanilla transformer-based bridge module that project the head node embedding to the space of the tail node using contrastive learning, though negative sampling (i.e., corrupting triples with randomly sampled negative tails). Consequently, BioBridge can transform an input embedding encoded using a unimodal FM to the space of the target modality according to their relation.

In Wang et al.[162], BioBridge has been evaluated for cross-modal retrieval, biomolecular functional similarity using the gene ontology (GO) annotations and protein-to-protein interaction tasks, outperforming baselines based on traditional KGE models, protein embedding models such as ProtBERT and ESM, and knowledge-guided protein embedding models like OntoProtein and KeAP. Across the baselines, for biomolecular functional similarity, methods augmented by KG – including KeAP and OntoProtein based on the ProteinKG25 dataset, containing approximately 4.8 million protein-GO triples – yield better results than the protein embedding models, implying that KG connecting proteins and the biological attributes enhance protein representation learning. For the protein-to-protein interaction, ESM2-3B performs better than KeAP, likely attributed to its pre-training on an extensive protein database, BioBRIDGE further enhances the embeddings using PrimeKG and demonstrates improved performance.

OtterKnowledge[160] learns a knowledge-enhanced representation by enriching the sequence/SMILES with MKGs fused from different sources. Initially, an MKG is constructed from the original datasets based on a user-provided domain schema. In Hoang et al.[160], four MKGs curated and integrated from Uniprot+BindingDB+ChEMBL, DUD-e, PrimeKG and STITCH are introduced . These MKGs are described as directed labeled graphs capturing structured knowledge, in the form of RDF triples: (subject, predicate, object), between entities of interest (e.g., proteins, small molecules, diseases, measurements, etc.).

An entity can connect to multiple attributes nodes with different modalities, each providing additional information about the entity (e.g., for a protein that can be its sequence, a description of its function, a 3D image, etc.), and to other entities through different semantic relationships (e.g., binds, interact, etc.). Then, a pretrained model is assigned to compute initial embedding for each modality. For example, models such as ESM 1B[35], morgan-fingerprint[19] or MolFormer[18] and BERT are used for protein sequences, molecules SMILES and textual properties, respectively. Subsequently, an inductive GNN trains on the MKG. First, a projection layer transforms each node modality into a common dimensionality using linear transformations. Then, the initial embeddings are propagated through several multi-relational graph convolutional layers (R-GCN) that enrich the input embeddings according to the node neighbours. R-GCN[165] distinguished between different types of edges by having trainable parameters for each edge type. The model is trained with multiple objectives, such link prediction or regression for numerical attributes. During inference, pretrained knowledge-enhanced GNN embeddings for drugs/proteins are acquired for the downstream tasks. These optimized embeddings and/or the GNN trained projection layers are then fed into downstream prediction tasks to improve drug-target binding affinity prediction.

In Hoang et al.[160], OtterKnowledge surpasses state-of-the-art results on three Therapeutic Data Commons (TDC) benchmarks[166] for drug-target binding affinity prediction: DTI DG, DAVIS and KIBA. The TDC DTI dataset's temporal split allows evaluating method generalization, since most of the molecules on test data are unseen in the training data (close to a real discovery scenario where you have new drug molecules). To address the issues of missing modalities, attribute modalities (such as a protein function or protein sequence) are treated as relational triples instead of predetermined features of an entity (e.g., protein) during training. This approach offers flexibility in handling multiple modalities for the same entity node, avoiding the need for nodes to carry all multimodal properties or project large property vectors with missing values. Instead, the projection is done per modality and only when such a modality exists for the entity[160].

8 Opportunities and open challenges

To advance life science discovery, democratizing the vast human knowledge accumulated in human-curated multi-modal bases or even ingested from scientific literature[167] is a key step. The size of the models matters, which requires developing methods capable of integrating a broader array of data sources into various models, including single-modal machine learning, multi-modal fusion techniques, and multi-modal knowledge graph methods, to take advantage of heterogeneous knowledge bases, scientific literature, clinical data and experimental data derived from simulations. These modalities need to be accessible in a consistent machine-readable form, and serve as a foundation for AI-enriched models for prediction, simulation, generative modelling and synthesis. To foster research in this topic, some insights into the research challenges and opportunities are provided here, aiming to enhance the discovery process by potentially advancing the state of the art for machine learning aided drug discovery.

8.1 Reusing traditional QSAR descriptors for deep learning tasks.

1D and 2D modalities such as SMILES and 2D fingerprints used traditionally by QSAR (quantitative structure-activity relationship) methods have been widely adapted for deep learning algorithms as discussed above. However, 3D or higher dimensional descriptors such as CoMFA[168], CoMSIA[169], G-WHIM[170], and VolSurf[171] are much less common in deep learning related research, with the exception of a handful of studies (that are significantly less cited than 3D graphs and 3D fingerprints)[72, 172-173]. We suspect that this is due to a "knowledge barrier" between traditional QSAR field and the emerging deep learning field. This knowledge barrier is perhaps caused by software behind paywalls, documentations that are either hard to access or domain-specific, or simply less than ideal communication between two fields (which is doubtful in 2024). Nonetheless, it is foreseeable that this issue will be resolved by the current active research field very soon.

8.2 Open question about the best practice of using MD trajectory in machine learning tasks.

4D-QSAR[174] and 4D modalities for machine learning tasks suffer from a similar problem – datasets that contains multiple molecular conformations are relatively rare[175-177]. Therefore, MD simulations (along with other simulations) provide an ideal surrogate to supplement the data scarcity. As discussed above, currently time-dependent machine learning mostly focuses on clustering and MD features learning. As for predictive tasks, CASTELO[116-117] uses comparative metrics between clustering results and predicts hot spots in molecules using flexibility information of molecules from MD trajectory. MD-Graph[118] uses GCN to process each individual frames before aggregating the results for immunogenicity predictions. It is clear that both these methods can be categorized as "late fusion" method assuming that 4D modality learning is a multimodal fusion problem. Therefore, it would be interesting to explore early fusion and intermediate fusion strategies for 4D modalities. Additionally, it is relatively time-consuming to collect MD trajectories. Building MD trajectory databases[178-179] will be helpful to further the development of machine learning with 4D modalities.

8.3 Incorporation of knowledge in existent models demands more efficient and scalable frameworks (KG).

The challenge lies in extracting and encoding information fused from many sources into vector representations, and effectively injecting large knowledge into machine learning and large language models (LLMs), which may help in turn to probe or provide provenance to the output of the LLMs[180]. This requires the implementation of efficient and scalable frameworks for training and inference, capable of handling large sources of knowledge, both in terms of size (i.e., number of triples) and heterogeneity (i.e., vocabulary scale to distinct relations and attributes). The management of KG storage, including KG evolution and maintenance, is an active area of research[181] and their performance has been object of systematic studies, also for biomedical uses cases[182]. However, there is a lack of graph stores that natively support different modalities and/or computation of embeddings based on that modality. Large MKGs are challenging for all embedding-based link prediction techniques, multimodal embeddings are not significantly worse as they are treated as additional triples. Nevertheless, multimodal encoder/decoders are more costly to train[161]. Techniques for batching, partitioning and sampling are usually required for training, for example, in Otter knowledge, GAS methods[183] are employed to scale the training.

8.4 Exploring multimodal research beyond language and vision to enable the discovery of new knowledge in scientific domains (KG).

While research in MKGs has predominantly centered on language (text) and vision (images), there is a need to delve into multimodal research across diverse modalities and domains, in particular for drug discovery. In addition, this might involve multi-task training with multiple objectives (e.g., link prediction, regression for numerical values) and the implementation of multimodal imputation models capable of generating missing multimodal values across a broader spectrum of modalities, such as generation of textual properties, like as protein function, a protein sequence or an image. This requires not just to combine encoders to learn embeddings of multimodal types used to predict the links, but also neural decoders to generate missing multimodal attributes from the information in the KG[161].

8.5 Generalizing learned representations to multiple downstream tasks.

Generalizing learned representations to multiple downstream tasks involves developing robust training techniques that enable predictions for entities with unseen or missing modalities (common in a discovery scenario). This includes transferring learned embeddings from pretrained models to multiple downstream tasks, and addressing the challenge of utilising multimodal triples where not all information is necessarily informative for all prediction downstream tasks[160-161]. One crucial aspect involves analysing how the disparity between the data accessible during pretraining and the data accessible during subsequent tasks may impact downstream tasks[160]. During training, numerous (multimodal) attributes may be associated with proteins or drugs, whereas during downstream fine-tuning, one may need to infer properties of proteins or sequences for which only amino acid sequences and SMILES are available[160].

8.6 The challenge of 1+1<2 faced by multimodal fusion problems.

If the aim is to achieve additive predictive power in multimodal fusion problems, each modality supplied must be orthogonal in its information space. This is, in reality, almost impossible to achieve, with an easy counterexample of AlphaFold which predicts 3D protein structures with 1D protein sequence. In other words, 1D protein sequences shares adequate amount of information with 3D protein structures for AlphaFold to succeed. However, this correlation is not obvious to most machine learning algorithms, which results in a frequently-seen "slight increment" in predictive power. To further boost the predictive power, an important question arises: how to provide as much information as possible, therefore removing the parallelization of information among different modalities? Assuming that all possible modalities have been exhausted to describe the protein-ligand complexes, the solution becomes mechanical – which to optimize a target value by selecting the correct combination of modalities and selecting a compatible machine learning algorithm. Such is the original goal of this chapter – to reflect on the question if we have exhausted descriptors. If so, it is vital to create an algorithm to select the best of the modalities for the task. If not, it is important to continue to explore important representations that might have been overlooked.

8.7 Handling long tail phenomenon and modality collapse.

This challenge arises from imbalances in training data and labels exhibiting a long-tailed distribution, i.e., a small ratio of labels are common, with a large number of training samples, while most labels are infrequent or even have never appeared before[140]. For example, in training data for rare diseases that affect only a small portion of the population. In addition, structure modalities for molecules or proteins can provide valuable insights for representation learning, but the sparsity of certain modalities, like 3D images, may lead to relatively small gains. A potential issue is modality collapse, wherein only a subset of the most helpful modalities dominates the training process in multimodal fusion problems and KG problems, leading to over-reliance and neglect of informative data from other modalities. This imbalance in the learning process or insufficient data for one or more modalities can lead to sub-optimal representations[124]. Moreover, biases may also be introduced due to poor quality of data in multimodal fusion problems and the sparseness and incompleteness of KGs, even when collected from multiple sources.

8.8 Learning representation across unaligned heterogeneous datasets.

Catastrophic forgetting may occur when, for example, training a model on datasets with slightly different schemas one at a time, the model may forget everything it learns in the previous database while learning from the new one. Alignment between source schemas is not a trivial problem, even if a relation in one data source may share some similarity with another in a different data source, since they are not exactly the same it is not possible to treat them as the same type of relation[160]. This hinders the model from effectively transferring commonalities between these two relations without an explicit mechanism to force the model to do so. To address this, OtterKnowledge employs ensemble methods to cope with pretrained models trained in unaligned KG separately[160]. However, an ensemble approach is not practical as the number of models that are needed is growing with the number of databases considered. Designing a dynamic learning approach such that the model can learn to transfer across data sources with different schemas is an open research question.

8.9 Benchmarking pretrained models and explainability.

This highlights the need of publicly available benchmarks and leaderboards tailored for assessing the effectiveness of pretrained models using diverse modalities[160]. Without standardized benchmarks, it becomes difficult to objectively evaluate the performance of cross-modal models compare to existing, single-modal models across for various tasks. Although datasets such as MoleculeNet[24], ChEMBL[121], DUD-e[184], DrugBank[120], and BindingDB[122] have been used as benchmarks by studies reviewed in this chapter, dataset biases have been found to cause model biases, leading to misleading results.[185-186] Ongoing database maintenance is of great importance along with funding long-lasting competitions such as CASP[187], CAPRI[188], CAFA[189], and D3R[190]. Additionally, the lack of explainability on the outputs of the models limits their practical applicability. Attention-based GNN[87-88] has been utilized to enhance the interpretability of molecule property predictions. For KG, an interesting research avenue is to explore how KG embedding models and neuro-symbolic methods can enhance the interpretability or understanding of the reasoning behind the model's decisions[191].

9 Conclusion

The surveyed methods are summarized in **Table 1** using the updated definitions of data modalities. It is important to re-emphasize the underlying physical meaning of these categories, which may imply orthogonality of information between different representations that is needed when selecting representations for multimodal fusion algorithms. Methods listed in this table extract information ranging from chemistry (chemical bonds, angles and dihedral angles, hydrogen bonds, atomic numbers, etc.), physics (electrostatics, hydrophobicity, quantum mechanics, etc.), topology (distances, manifolds, point groups, etc.), biology (biological functions, gene ontology, gene evolution, etc.). However, it is likely that molecular information of molecules have not been exhausted by current methods. With the fast progress in the field of machine learning aided drug discovery, it is anticipated to have new additions to each category and even additions of new categories in the future.

Table 1 Summary of methods reviewed in this chapter.

Main category	Subcategory	Methods
Sequential modalities	1D sequences	Protein sequence, SMILES[16], SMARTS[21], SELFIES[22], InChI[25], HELM[40], CHUCKLES[41], HPNN[42]
Topological modalities	2D fingerprints	MACCS[45], Daylight[46], ECFP[47], FCFP[47]
	2D images	Chemception[52], Meyer et al.[53], DEEPScreen[54]
	2D graphs	AquaSol[89], Weave[92]
Spatial modalities	Distance/contact matrices	Distogram (AlphaFold)[33], RF-Score[59], HIP-NN[58], OnionNet[60]

	3D fingerprints	NNScore[61], ECIF[63], PLIF[65], SPLIF[67], PLIP[68], ProLIF[69], E3FP[71], Khani et al.[72], Ji et al.[73], TopologyNet[74], DG-GL[75], DC-GBT[76]
	3D images	Ragoza et al.[77], DeepSite[79], KDEEP[80], LigDream[81], Pafnuncy[82], MaSIF[83], PINet[84]
	3D graphs	DeepChem[94], SchNet[100], G-SchNet[101], 3D-Graph-SBDD[102], DimeNet[103], SpinConv[104], PhysNet[105], SphereNet[106], L-Net[97], DGraphDTA[96], HoloProt[99], RoseTTAFold[98], GemNet[90], DGANN[91]
Temporal modalities	4D	MD-IFP[112], Ribeiro et al.[113], CASTELO[116-117], MD-Graph[118]
Knowledge graphs	Knowledge graphs	OntoProtein[125], KeAP[127], ProtST[128], GOLabeler[131], DeepGraphGO[132], DeepGOZero[133], TxGNN[134] (PrimeKG[144])
Multimodal fusion methods	Early fusion[10]	Zhang et al.[153], GraphMVP[154], GEOSSL-DDM[155]
	Intermediate fusion[10]	MDeePred[156], CPAC[157], Jones et al.[159]
	Late fusion[10]	Jones et al.[159]
	Multimodal Knowledge graph	OtterKnowledge[160], BioBridge[162]

Table 2 Abbreviations used in this chapter

Abbreviation	Full name
1D	One-dimensional/Sequential
2D	Two-dimensional/Topological
3D	Three-dimensional/Spatial
4D	Four-dimensional/Temporal
AA	Amino acid
AI	Artificial intelligence
BERT	Bidirectional encoder representations from transformers
CAFA	Critical assessment of protein function annotation algorithms
CAPRI	Critical assessment of prediction of interactions
CASP	Critical assessment of structure prediction

CNN	Convolutional neural network
CoMFA	the Comparative molecular field analysis
COVID-19	Coronavirus disease 2019
CVAE	Convolutional variational autoencoder
D3R	Drug design data resource
DBN	Deep belief network
DNA	Deoxyribonucleic acid
DSM	Denoising score matching
DTI	Drug target interaction
ECFP	Extended-connectivity fingerprint
ECIF	Extended connectivity interaction features
ESM	Evolutionary scale modeling
FCFP	Functional-class fingerprint
FM	Foundation model
GAS	GNN auto scale
GAT	Graph attention network
GCN	Graph convolution network
GNN	Graph neural network
GO	Gene ontology
HDBSCAN	Hierarchical density-based spatial clustering of applications with noise
HLA	Human leukocyte antigen
HOMO	Highest occupied molecular orbital
IC50	Half-maximal inhibitory concentration
InChI	International Chemistry Identifier
KG	Knowledge graph
KGE	Knowledge graph embedding
LLM	Large language model
LUMO	Lowest unoccupied molecular orbital
MACCS	Molecular access System
MAT	Molecular attention network
MD	Molecular dynamics
MKG	Multimodal knowledge graph
ML	Machine learning
MM-GBSA	Molecular mechanics energies combined with generalized Born and surface area continuum solvation
MOE	Molecular operating environment (software)
MSA	Multiple sequence alignment
NLP	Natural language processing
NN	Neural network
PCA	Principal component analysis
PLIF	Protein-ligand interaction fingerprint
PLM	Protein language model
QSAR	Quantitative structure activity relationship

RBF	Radial basis function
RDF triple	Resource description framework triple
RF	Random forest
RGB	Red, green, and blue
RMSD	Root mean square displacement
RNA	Ribonucleic acid
RNN	Recursive neural network
SARS-CoV-2	Severe-acute-respiratory-syndrome-related coronavirus 2
SBDD	Structure-based drug discovery
SE	Special Euclidean group
SELFIES	Self-referencing embedded strings
SMARTS	SMILES arbitrary target specification
SMILES	Simplified Molecular Input Line Entry System
TIP3P	Three-site model (water model)

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References

1. Xiao, K.; Qian, Z.; Qin, B., A survey of data representation for multi-modality event detection and evolution. *Applied Sciences* **2022,** *12* (4), 2204.
2. Huang, K.; Xiao, C.; Glass, L. M.; Critchlow, C. W.; Gibson, G.; Sun, J., Machine learning applications for therapeutic tasks with genomics data. *Patterns* **2021,** *2* (10).
3. Todeschini, R. C., Viviana Molecular descriptors. In *Recent Advances in QSAR Studies*, 2010; pp 29-102.
4. Lo, Y. C.; Rensi, S. E.; Torng, W.; Altman, R. B., Machine learning in chemoinformatics and drug discovery. *Drug Discov Today* **2018,** *23* (8), 1538-1546.
5. Martinelli, D. D., Generative machine learning for de novo drug discovery: A systematic review. *Comput Biol Med* **2022,** *145*, 105403.
6. Askr, H.; Elgeldawi, E.; Aboul Ella, H.; Elshaier, Y.; Gomaa, M. M.; Hassanien, A. E., Deep learning in drug discovery: an integrative review and future challenges. *Artif Intell Rev* **2023,** *56* (7), 5975-6037.

7. Staszak, M.; Staszak, K.; Wieszczycka, K.; Bajek, A.; Roszkowski, K.; Tylkowski, B., Machine learning in drug design: Use of artificial intelligence to explore the chemical structure–biological activity relationship. *WIREs Computational Molecular Science* **2022,** *12* (2), e1568.
8. Floresta, G.; Zagni, C.; Gentile, D.; Patamia, V.; Rescifina, A., Artificial Intelligence Technologies for COVID-19 De Novo Drug Design. *Int J Mol Sci* **2022,** *23* (6).
9. Blanco-Gonzalez, A.; Cabezon, A.; Seco-Gonzalez, A.; Conde-Torres, D.; Antelo-Riveiro, P.; Pineiro, A.; Garcia-Fandino, R., The Role of AI in Drug Discovery: Challenges, Opportunities, and Strategies. *Pharmaceuticals (Basel)* **2023,** *16* (6).
10. Stahlschmidt, S. R.; Ulfenborg, B.; Synnergren, J., Multimodal deep learning for biomedical data fusion: a review. *Brief Bioinform* **2022,** *23* (2).
11. Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A. N.; Kaiser, Ł. U.; Polosukhin, I., Attention is All you Need. In *Advances in Neural Information Processing Systems*, Guyon, I.; Luxburg, U. V.; Bengio, S.; Wallach, H.; Fergus, R.; Vishwanathan, S.; Garnett, R., Eds. Curran Associates, Inc.: 2017; Vol. 30.
12. Radford, A. N., K.; Salimans, T.; Sutskever, I., Improving language understanding by generative pre-training. **2018**.
13. Devlin, J.; Chang, M.-W.; Lee, K.; Toutanova, K., BERT: Pre-training of deep bidirectional Transformers for language understanding. *arXiv [cs.CL]* **2018**.
14. Elnaggar, A.; Heinzinger, M.; Dallago, C.; Rehawi, G.; Wang, Y.; Jones, L.; Gibbs, T.; Feher, T.; Angerer, C.; Steinegger, M.; Bhowmik, D.; Rost, B., ProtTrans: Toward Understanding the Language of Life Through Self-Supervised Learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **2022,** *44* (10), 7112-7127.
15. Huang, K.; Xiao, C.; Glass, L. M.; Sun, J., MolTrans: Molecular Interaction Transformer for drug–target interaction prediction. *Bioinformatics* **2021,** *37* (6), 830-836.
16. Weininger, D., SMILES, a chemical language and information system. 1. Introduction to methodology and encoding rules. *Journal of Chemical Information and Computer Sciences* **1988,** *28* (1), 31-36.
17. Bjerrum, E. J.; Sattarov, B., Improving Chemical Autoencoder Latent Space and Molecular De Novo Generation Diversity with Heteroencoders. *Biomolecules* **2018,** *8* (4), 131.
18. Ross, J.; Belgodere, B.; Chenthamarakshan, V.; Padhi, I.; Mroueh, Y.; Das, P., Large-scale chemical language representations capture molecular structure and properties. *Nat Mach Intell* **2022,** *4* (12), 1256-1264.
19. Morgan, H. L., The Generation of a Unique Machine Description for Chemical Structures-A Technique Developed at Chemical Abstracts Service. *Journal of Chemical Documentation* **1965,** *5* (2), 107-113.
20. O'Boyle, N. M., Towards a Universal SMILES representation - A standard method to generate canonical SMILES based on the InChI. *Journal of Cheminformatics* **2012,** *4* (1), 22.
21. Chemaxon Chemaxon Extended SMILES and SMARTS - CXSMILES and CXSMARTS. https://docs.chemaxon.com/display/docs/chemaxon-extended-smiles-and-smarts-cxsmiles-and-cxsmarts.md.
22. Krenn, M.; Häse, F.; Nigam, A.; Friederich, P.; Aspuru-Guzik, A., Self-referencing embedded strings (SELFIES): A 100% robust molecular string representation. *Machine Learning: Science and Technology* **2020,** *1* (4), 045024.
23. Landrum, G., RDKit: A software suite for cheminformatics, computational chemistry, and predictive modeling. *Greg Landrum* **2013,** *8*, 31.
24. Wu, Z.; Ramsundar, B.; Feinberg, E. N.; Gomes, J.; Geniesse, C.; Pappu, A. S.; Leswing, K.; Pande, V., MoleculeNet: a benchmark for molecular machine learning. *Chemical science* **2018,** *9* (2), 513-530.
25. Heller, S. R.; McNaught, A.; Pletnev, I.; Stein, S.; Tchekhovskoi, D., InChI, the IUPAC International Chemical Identifier. *Journal of Cheminformatics* **2015,** *7* (1), 23.

26. Southan, C., InChI in the wild: an assessment of InChIKey searching in Google. *Journal of Cheminformatics* **2013,** *5* (1), 10.
27. Golts, A.; Ratner, V.; Shoshan, Y.; Raboh, M.; Polaczek, S.; Hexter, E., Large curated dataset for drug target interaction. Zenodo: 2023.
28. Samuel, S.; Rohit, S.; Lenore, C.; Bonnie, B., Adapting protein language models for rapid DTI prediction. *bioRxiv* **2022**, 2022.11.03.515084.
29. Israel, I. R.-. Fuse-drug example implementing PLM DTI model trained on a large DTI dataset. https://github.com/BiomedSciAI/fuse-drug/tree/main/fusedrug_examples/interaction/drug_target/affinity_prediction/PLM_DTI.
30. Golts, A.; Raboh, M.; Shoshan, Y.; Polaczek, S.; Rabinovici-Cohen, S.; Hexter, E., FuseMedML: a framework for accelerated discovery in machine learning based biomedicine. *J. Open Source Softw.* **2023,** *8* (81), 4943.
31. Chowdhury, B.; Garai, G., A review on multiple sequence alignment from the perspective of genetic algorithm. *Genomics* **2017,** *109* (5-6), 419-431.
32. Thompson, J. D.; Linard, B.; Lecompte, O.; Poch, O., A comprehensive benchmark study of multiple sequence alignment methods: current challenges and future perspectives. *PloS one* **2011,** *6* (3), e18093.
33. Jumper, J.; Evans, R.; Pritzel, A.; Green, T.; Figurnov, M.; Ronneberger, O.; Tunyasuvunakool, K.; Bates, R.; Žídek, A.; Potapenko, A.; Bridgland, A.; Meyer, C.; Kohl, S. A. A.; Ballard, A. J.; Cowie, A.; Romera-Paredes, B.; Nikolov, S.; Jain, R.; Adler, J.; Back, T.; Petersen, S.; Reiman, D.; Clancy, E.; Zielinski, M.; Steinegger, M.; Pacholska, M.; Berghammer, T.; Bodenstein, S.; Silver, D.; Vinyals, O.; Senior, A. W.; Kavukcuoglu, K.; Kohli, P.; Hassabis, D., Highly accurate protein structure prediction with AlphaFold. *Nature* **2021,** *596* (7873), 583-589.
34. Fernández-Díaz, R.; Hoang, T. L.; Lopez, V.; Shields, D. C., Effect of dataset partitioning strategies for evaluating out-of-distribution generalisation for predictive models in biochemistry. *bioRxiv* **2024**, 2024.03. 14.584508.
35. Rives, A.; Meier, J.; Sercu, T.; Goyal, S.; Lin, Z.; Liu, J.; Guo, D.; Ott, M.; Zitnick, C. L.; Ma, J.; Fergus, R., Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proceedings of the National Academy of Sciences* **2021,** *118* (15), e2016239118.
36. Brandes, N.; Ofer, D.; Peleg, Y.; Rappoport, N.; Linial, M., ProteinBERT: a universal deep-learning model of protein sequence and function. *Bioinformatics* **2022,** *38* (8), 2102-2110.
37. Rao, R.; Bhattacharya, N.; Thomas, N.; Duan, Y.; Chen, P.; Canny, J.; Abbeel, P.; Song, Y. In *Evaluating Protein Transfer Learning with TAPE*, 2019; Curran Associates, Inc.: 2019.
38. Born, J.; Shoshan, Y.; Huynh, T.; Cornell, W. D.; Martin, E. J.; Manica, M., On the Choice of Active Site Sequences for Kinase-Ligand Affinity Prediction. *Journal of Chemical Information and Modeling* **2022,** *62* (18), 4295-4299.
39. Chen, W. L.; Leland, B. A.; Durant, J. L.; Grier, D. L.; Christie, B. D.; Nourse, J. G.; Taylor, K. T., Self-Contained Sequence Representation: Bridging the Gap between Bioinformatics and Cheminformatics. *Journal of Chemical Information and Modeling* **2011,** *51* (9), 2186-2208.
40. Zhang, T.; Li, H.; Xi, H.; Stanton, R. V.; Rotstein, S. H., HELM: A Hierarchical Notation Language for Complex Biomolecule Structure Representation. *Journal of Chemical Information and Modeling* **2012,** *52* (10), 2796-2806.
41. Siani, M. A.; Weininger, D.; Blaney, J. M., CHUCKLES: A method for representing and searching peptide and peptoid sequences on both monomer and atomic levels. *Journal of Chemical Information and Computer Sciences* **1994,** *34* (3), 588-593.
42. Karimi, M.; Wu, D.; Wang, Z.; Shen, Y., Explainable Deep Relational Networks for Predicting Compound–Protein Affinities and Contacts. *Journal of Chemical Information and Modeling* **2021,** *61* (1), 46-66.
43. David, L.; Thakkar, A.; Mercado, R.; Engkvist, O., Molecular representations in AI-driven drug discovery: a review and practical guide. *Journal of Cheminformatics* **2020,** *12* (1), 56.

44. Xue, L.; Bajorath, J., Molecular descriptors in chemoinformatics, computational combinatorial chemistry, and virtual screening. *Combinatorial chemistry & high throughput screening* **2000,** *3* (5), 363-372.
45. Durant, J. L.; Leland, B. A.; Henry, D. R.; Nourse, J. G., Reoptimization of MDL keys for use in drug discovery. *Journal of chemical information and computer sciences* **2002,** *42* (6), 1273-1280.
46. Daylight Fingerprints. https://www.daylight.com/meetings/summerschool01/course/basics/fp.html.
47. Rogers, D.; Hahn, M., Extended-connectivity fingerprints. *Journal of chemical information and modeling* **2010,** *50* (5), 742-754.
48. Krizhevsky, A.; Sutskever, I.; Hinton, G. E., ImageNet Classification with Deep Convolutional Neural Networks. In *Advances in Neural Information Processing Systems*, Pereira, F.; Burges, C. J.; Bottou, L.; Weinberger, K. Q., Eds. Curran Associates, Inc.: 2012; Vol. 25.
49. Girshick, R., Fast R-CNN. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2015.
50. He, K.; Zhang, X.; Ren, S.; Sun, J., Deep Residual Learning for Image Recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
51. Dosovitskiy, A.; Beyer, L.; Kolesnikov, A.; Weissenborn, D.; Zhai, X.; Unterthiner, T.; Dehghani, M.; Minderer, M.; Heigold, G.; Gelly, S.; Uszkoreit, J.; Houlsby, N., An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. *arXiv [cs.CV]* **2021**.
52. Goh, G. B.; Siegel, C.; Vishnu, A.; Hodas, N. O.; Baker, N., Chemception: A deep neural network with minimal chemistry knowledge matches the performance of expert-developed QSAR/QSPR models. *arXiv [stat.ML]* **2017**.
53. Meyer, J. G.; Liu, S.; Miller, I. J.; Coon, J. J.; Gitter, A., Learning Drug Functions from Chemical Structures with Convolutional Neural Networks and Random Forests. *Journal of Chemical Information and Modeling* **2019,** *59* (10), 4438-4449.
54. Rifaioglu, A. S.; Nalbat, E.; Atalay, V.; Martin, M. J.; Cetin-Atalay, R.; Doğan, T., DEEPScreen: high performance drug-target interaction prediction with convolutional neural networks using 2-D structural compound representations. *Chem. Sci.* **2020,** *11* (9), 2531-2557.
55. Zeng, X.; Xiang, H.; Yu, L.; Wang, J.; Li, K.; Nussinov, R.; Cheng, F., Accurate prediction of molecular properties and drug targets using a self-supervised image representation learning framework. *Nat Mach Intell* **2022,** *4* (11), 1004-1016.
56. Selvaraju, R. R.; Cogswell, M.; Das, A.; Vedantam, R.; Parikh, D.; Batra, D. In *Grad-cam: Visual explanations from deep networks via gradient-based localization*, Proceedings of the IEEE international conference on computer vision, 2017; pp 618-626.
57. Fuchs, F.; Worrall, D.; Fischer, V.; Welling, M., Se (3)-transformers: 3d roto-translation equivariant attention networks. *Advances in neural information processing systems* **2020,** *33*, 1970-1981.
58. Lubbers, N.; Smith, J. S.; Barros, K., Hierarchical modeling of molecular energies using a deep neural network. *The Journal of chemical physics* **2018,** *148* (24).
59. Ballester, P. J.; Mitchell, J. B., A machine learning approach to predicting protein–ligand binding affinity with applications to molecular docking. *Bioinformatics* **2010,** *26* (9), 1169-1175.
60. Zheng, L.; Fan, J.; Mu, Y., Onionnet: a multiple-layer intermolecular-contact-based convolutional neural network for protein–ligand binding affinity prediction. *ACS omega* **2019,** *4* (14), 15956-15965.
61. Durrant, J. D.; McCammon, J. A., NNScore: a neural-network-based scoring function for the characterization of protein− ligand complexes. *Journal of chemical information and modeling* **2010,** *50* (10), 1865-1871.
62. Trott, O.; Olson, A. J., AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of computational chemistry* **2010,** *31* (2), 455-461.
63. Sánchez-Cruz, N.; Medina-Franco, J. L.; Mestres, J.; Barril, X., Extended connectivity interaction features: improving binding affinity prediction through chemical description. *Bioinformatics* **2021,** *37* (10), 1376-1382.

64. Landrum, G., RDKit: A software suite for cheminformatics, computational chemistry, and predictive modeling. *rdkit.org* **2013,** *8*.
65. Vilar, S.; Cozza, G.; Moro, S., Medicinal chemistry and the molecular operating environment (MOE): application of QSAR and molecular docking to drug discovery. *Current topics in medicinal chemistry* **2008,** *8* (18), 1555-1572.
66. Rosell-Hidalgo, A.; Young, L.; Moore, A. L.; Ghafourian, T., QSAR and molecular docking for the search of AOX inhibitors: a rational drug discovery approach. *Journal of Computer-Aided Molecular Design* **2021,** *35*, 245-260.
67. Da, C.; Kireev, D., Structural protein–ligand interaction fingerprints (SPLIF) for structure-based virtual screening: method and benchmark study. *Journal of chemical information and modeling* **2014,** *54* (9), 2555-2561.
68. Salentin, S.; Schreiber, S.; Haupt, V. J.; Adasme, M. F.; Schroeder, M., PLIP: fully automated protein–ligand interaction profiler. *Nucleic acids research* **2015,** *43* (W1), W443-W447.
69. Bouysset, C.; Fiorucci, S., ProLIF: a library to encode molecular interactions as fingerprints. *Journal of cheminformatics* **2021,** *13*, 1-9.
70. Michaud-Agrawal, N.; Denning, E. J.; Woolf, T. B.; Beckstein, O., MDAnalysis: a toolkit for the analysis of molecular dynamics simulations. *Journal of computational chemistry* **2011,** *32* (10), 2319-2327.
71. Axen, S. D.; Huang, X.-P.; Cáceres, E. L.; Gendelev, L.; Roth, B. L.; Keiser, M. J., A simple representation of three-dimensional molecular structure. *Journal of medicinal chemistry* **2017,** *60* (17), 7393-7409.
72. Khani, H.; Sepehrifar, M. B.; Yarahmadian, S., An improvement on the prediction power of the 3D-QSAR CoMFA models using a hybrid of statistical and machine learning methods: a case study on γ-secretase modulators of Alzheimer's disease. *Medicinal Chemistry Research* **2017,** *26* (6), 1184-1200.
73. Ji, B.; He, X.; Zhai, J.; Zhang, Y.; Man, V. H.; Wang, J., Machine learning on ligand-residue interaction profiles to significantly improve binding affinity prediction. *Briefings in Bioinformatics* **2021,** *22* (5).
74. Cang, Z.; Wei, G.-W., TopologyNet: Topology based deep convolutional and multi-task neural networks for biomolecular property predictions. *PLoS computational biology* **2017,** *13* (7), e1005690.
75. Nguyen, D. D.; Wei, G. W., DG-GL: Differential geometry-based geometric learning of molecular datasets. *Int J Numer Method Biomed Eng* **2019,** *35* (3), e3179.
76. Liu, X.; Feng, H.; Wu, J.; Xia, K., Dowker complex based machine learning (DCML) models for protein-ligand binding affinity prediction. *PLoS Comput Biol* **2022,** *18* (4), e1009943.
77. Ragoza, M.; Hochuli, J.; Idrobo, E.; Sunseri, J.; Koes, D. R., Protein–ligand scoring with convolutional neural networks. *Journal of chemical information and modeling* **2017,** *57* (4), 942-957.
78. Koes, D. R.; Baumgartner, M. P.; Camacho, C. J., Lessons learned in empirical scoring with smina from the CSAR 2011 benchmarking exercise. *Journal of chemical information and modeling* **2013,** *53* (8), 1893-1904.
79. Jimenez, J.; Doerr, S.; Martinez-Rosell, G.; Rose, A. S.; De Fabritiis, G., DeepSite: protein-binding site predictor using 3D-convolutional neural networks. *Bioinformatics* **2017,** *33* (19), 3036-3042.
80. Jiménez, J.; Škalič, M.; Martínez-Rosell, G.; De Fabritiis, G., KDEEP: Protein–Ligand Absolute Binding Affinity Prediction via 3D-Convolutional Neural Networks. *Journal of Chemical Information and Modeling* **2018,** *58* (2), 287-296.
81. Skalic, M.; Jiménez, J.; Sabbadin, D.; De Fabritiis, G., Shape-Based Generative Modeling for de Novo Drug Design. *Journal of Chemical Information and Modeling* **2019,** *59* (3), 1205-1214.
82. Stepniewska-Dziubinska, M. M.; Zielenkiewicz, P.; Siedlecki, P., Development and evaluation of a deep learning model for protein–ligand binding affinity prediction. *Bioinformatics* **2018,** *34* (21), 3666-3674.
83. Gainza, P.; Sverrisson, F.; Monti, F.; Rodola, E.; Boscaini, D.; Bronstein, M. M.; Correia, B. E., Deciphering interaction fingerprints from protein molecular surfaces using geometric deep learning. *Nat Methods* **2020,** *17* (2), 184-192.

84. Dai, B.; Bailey-Kellogg, C., Protein interaction interface region prediction by geometric deep learning. *Bioinformatics* **2021,** *37* (17), 2580-2588.
85. Scarselli, F.; Gori, M.; Tsoi, A. C.; Hagenbuchner, M.; Monfardini, G., The graph neural network model. *IEEE transactions on neural networks* **2008,** *20* (1), 61-80.
86. Zhou, J.; Cui, G.; Hu, S.; Zhang, Z.; Yang, C.; Liu, Z.; Wang, L.; Li, C.; Sun, M., Graph neural networks: A review of methods and applications. *AI open* **2020,** *1*, 57-81.
87. Velickovic, P.; Cucurull, G.; Casanova, A.; Romero, A.; Lio, P.; Bengio, Y., Graph attention networks. *stat* **2017,** *1050* (20), 10-48550.
88. Maziarka, Ł.; Danel, T.; Mucha, S.; Rataj, K.; Tabor, J.; Jastrzębski, S., Molecule Attention Transformer. *arXiv [cs.LG]* **2020**.
89. Lusci, A.; Pollastri, G.; Baldi, P., Deep architectures and deep learning in chemoinformatics: the prediction of aqueous solubility for drug-like molecules. *Journal of chemical information and modeling* **2013,** *53* (7), 1563-1575.
90. Gasteiger, J.; Becker, F.; Günnemann, S., Gemnet: Universal directional graph neural networks for molecules. *Advances in Neural Information Processing Systems* **2021,** *34*, 6790-6802.
91. Qian, C.; Xiong, Y.; Chen, X., Directed graph attention neural network utilizing 3d coordinates for molecular property prediction. *Computational Materials Science* **2021,** *200*, 110761.
92. Kearnes, S.; McCloskey, K.; Berndl, M.; Pande, V.; Riley, P., Molecular graph convolutions: moving beyond fingerprints. *Journal of computer-aided molecular design* **2016,** *30*, 595-608.
93. Kipf, T. N.; Welling, M., Semi-Supervised Classification with Graph Convolutional Networks. *arXiv [cs.LG]* **2017**.
94. Altae-Tran, H.; Ramsundar, B.; Pappu, A. S.; Pande, V., Low data drug discovery with one-shot learning. *ACS central science* **2017,** *3* (4), 283-293.
95. Morrone, J. A.; Weber, J. K.; Huynh, T.; Luo, H.; Cornell, W. D., Combining docking pose rank and structure with deep learning improves protein–ligand binding mode prediction over a baseline docking approach. *Journal of chemical information and modeling* **2020,** *60* (9), 4170-4179.
96. Jiang, M.; Li, Z.; Zhang, S.; Wang, S.; Wang, X.; Yuan, Q.; Wei, Z., Drug–target affinity prediction using graph neural network and contact maps. *RSC advances* **2020,** *10* (35), 20701-20712.
97. Li, Y.; Pei, J.; Lai, L., Structure-based de novo drug design using 3D deep generative models. *Chemical science* **2021,** *12* (41), 13664-13675.
98. Baek, M.; DiMaio, F.; Anishchenko, I.; Dauparas, J.; Ovchinnikov, S.; Lee, G. R.; Wang, J.; Cong, Q.; Kinch, L. N.; Schaeffer, R. D., Accurate prediction of protein structures and interactions using a three-track neural network. *Science* **2021,** *373* (6557), 871-876.
99. Somnath, V. R.; Bunne, C.; Krause, A., Multi-scale representation learning on proteins. *Advances in Neural Information Processing Systems* **2021,** *34*, 25244-25255.
100. Schütt, K.; Kindermans, P.-J.; Sauceda Felix, H. E.; Chmiela, S.; Tkatchenko, A.; Müller, K.-R., Schnet: A continuous-filter convolutional neural network for modeling quantum interactions. *Advances in neural information processing systems* **2017,** *30*.
101. Gebauer, N.; Gastegger, M.; Schütt, K., Symmetry-adapted generation of 3d point sets for the targeted discovery of molecules. *Advances in neural information processing systems* **2019,** *32*.
102. Luo, S.; Guan, J.; Ma, J.; Peng, J., A 3D generative model for structure-based drug design. *Advances in Neural Information Processing Systems* **2021,** *34*, 6229-6239.
103. Gasteiger, J.; Groß, J.; Günnemann, S., Directional message passing for molecular graphs. *arXiv preprint arXiv:2003.03123* **2020**.
104. Shuaibi, M.; Kolluru, A.; Das, A.; Grover, A.; Sriram, A.; Ulissi, Z.; Zitnick, C. L., Rotation invariant graph neural networks using spin convolutions. *arXiv preprint arXiv:2106.09575* **2021**.
105. Unke, O. T.; Meuwly, M., PhysNet: A neural network for predicting energies, forces, dipole moments, and partial charges. *Journal of chemical theory and computation* **2019,** *15* (6), 3678-3693.
106. Liu, Y.; Wang, L.; Liu, M.; Lin, Y.; Zhang, X.; Oztekin, B.; Ji, S. In *Spherical message passing for 3d molecular graphs*, International Conference on Learning Representations (ICLR), 2022.

107. Baillif, B.; Cole, J.; McCabe, P.; Bender, A., Deep generative models for 3D molecular structure. *Current Opinion in Structural Biology* **2023,** *80*, 102566.
108. Isert, C.; Atz, K.; Schneider, G., Structure-based drug design with geometric deep learning. *Current Opinion in Structural Biology* **2023,** *79*, 102548.
109. Sittel, F.; Jain, A.; Stock, G., Principal component analysis of molecular dynamics: on the use of Cartesian vs. internal coordinates. *J Chem Phys* **2014,** *141* (1), 014111.
110. Salo-Ahen, O. M. H.; Alanko, I.; Bhadane, R.; Bonvin, A. M. J. J.; Honorato, R. V.; Hossain, S.; Juffer, A. H.; Kabedev, A.; Lahtela-Kakkonen, M.; Larsen, A. S.; Lescrinier, E.; Marimuthu, P.; Mirza, M. U.; Mustafa, G.; Nunes-Alves, A.; Pantsar, T.; Saadabadi, A.; Singaravelu, K.; Vanmeert, M. Molecular Dynamics Simulations in Drug Discovery and Pharmaceutical Development *Processes* [Online], 2021.
111. Xia, S.; Chen, E.; Zhang, Y., Integrated Molecular Modeling and Machine Learning for Drug Design. *Journal of Chemical Theory and Computation* **2023,** *19* (21), 7478-7495.
112. Kokh, D. B.; Doser, B.; Richter, S.; Ormersbach, F.; Cheng, X.; Wade, R. C., A workflow for exploring ligand dissociation from a macromolecule: Efficient random acceleration molecular dynamics simulation and interaction fingerprint analysis of ligand trajectories. *J Chem Phys* **2020,** *153* (12), 125102.
113. Lamim Ribeiro, J. M.; Provasi, D.; Filizola, M., A combination of machine learning and infrequent metadynamics to efficiently predict kinetic rates, transition states, and molecular determinants of drug dissociation from G protein-coupled receptors. *J Chem Phys* **2020,** *153* (12), 124105.
114. Ribeiro, J. M. L.; Bravo, P.; Wang, Y.; Tiwary, P., Reweighted autoencoded variational Bayes for enhanced sampling (RAVE). *J Chem Phys* **2018,** *149* (7), 072301.
115. Ravindra, P.; Smith, Z.; Tiwary, P., Automatic mutual information noise omission (AMINO): generating order parameters for molecular systems. *Mol. Syst. Des. Eng.* **2020,** *5* (1), 339-348.
116. Zhang, L.; Domeniconi, G.; Yang, C.-C.; Kang, S.-g.; Zhou, R.; Cong, G., CASTELO: clustered atom subtypes aided lead optimization—a combined machine learning and molecular modeling method. *BMC Bioinform.* **2021,** *22*, 1-21.
117. Bell, D. R.; Domeniconi, G.; Yang, C.-C.; Zhou, R.; Zhang, L.; Cong, G., Dynamics-Based Peptide–MHC Binding Optimization by a Convolutional Variational Autoencoder: A Use-Case Model for CASTELO. *Journal of Chemical Theory and Computation* **2021,** *17* (12), 7962-7971.
118. Weber, J. K. e. a., Deep learning on molecular simulations reveals dynamical determinants of HLA-peptide complex immunogenicity. *Briefings in Bioinformatics* **2024,** (Just accepted @@@).
119. The UniProt, C., UniProt: a worldwide hub of protein knowledge. *Nucleic Acids Research* **2019,** *47* (D1), D506-D515.
120. Law, V.; Knox, C.; Djoumbou, Y.; Jewison, T.; Guo, A. C.; Liu, Y.; Maciejewski, A.; Arndt, D.; Wilson, M.; Neveu, V.; Tang, A.; Gabriel, G.; Ly, C.; Adamjee, S.; Dame, Z. T.; Han, B.; Zhou, Y.; Wishart, D. S., DrugBank 4.0: shedding new light on drug metabolism. *Nucleic Acids Research* **2013,** *42* (D1), D1091-D1097.
121. Gaulton, A.; Bellis, L. J.; Bento, A. P.; Chambers, J.; Davies, M.; Hersey, A.; Light, Y.; McGlinchey, S.; Michalovich, D.; Al-Lazikani, B.; Overington, J. P., ChEMBL: a large-scale bioactivity database for drug discovery. *Nucleic Acids Research* **2012,** *40* (D1), D1100-D1107.
122. Liu, T.; Lin, Y.; Wen, X.; Jorissen, R. N.; Gilson, M. K., BindingDB: a web-accessible database of experimentally determined protein–ligand binding affinities. *Nucleic Acids Research* **2007,** *35* (suppl_1), D198-D201.
123. Barbarino, J. M.; Whirl-Carrillo, M.; Altman, R. B.; Klein, T. E., PharmGKB: a worldwide resource for pharmacogenomic information. *Wiley Interdisciplinary Reviews: Systems Biology and Medicine* **2018,** *10* (4), e1417.
124. Ektefaie, Y.; Dasoulas, G.; Noori, A.; Farhat, M.; Zitnik, M., Multimodal learning with graphs. *Nat Mach Intell* **2023,** *5* (4), 340-350.
125. Zhang, N.; Bi, Z.; Liang, X.; Cheng, S.; Hong, H.; Deng, S.; Lian, J.; Zhang, Q.; Chen, H. In *OntoProtein: Protein Pretraining With Gene Ontology Embedding*, 2022/01/23/; 2022.

126. Ashburner, M.; Ball, C. A.; Blake, J. A.; Botstein, D.; Butler, H.; Cherry, J. M.; Davis, A. P.; Dolinski, K.; Dwight, S. S.; Eppig, J. T., Gene ontology: tool for the unification of biology. *Nature genetics* **2000,** *25* (1), 25-29.
127. Zhou, H.-Y.; Fu, Y.; Zhang, Z.; Cheng, B.; Yu, Y. In *Protein Representation Learning via Knowledge Enhanced Primary Structure Reasoning*, The Eleventh International Conference on Learning Representations, 2023/02/01/; 2023.
128. Xu, M.; Yuan, X.; Miret, S.; Tang, J., ProtST: Multi-Modality Learning of Protein Sequences and Biomedical Texts. arXiv: 2023.
129. Zhao, Y.; Wang, J.; Chen, J.; Zhang, X.; Guo, M.; Yu, G., A Literature Review of Gene Function Prediction by Modeling Gene Ontology. *Frontiers in Genetics* **2020,** *11*.
130. Unsal, S.; Atas, H.; Albayrak, M.; Turhan, K.; Acar, A. C.; Doğan, T., Learning functional properties of proteins with language models. *Nat Mach Intell* **2022,** *4* (3), 227-245.
131. You, R.; Zhang, Z.; Xiong, Y.; Sun, F.; Mamitsuka, H.; Zhu, S., GOLabeler: improving sequence-based large-scale protein function prediction by learning to rank. *Bioinformatics* **2018,** *34* (14), 2465-2473.
132. You, R.; Yao, S.; Mamitsuka, H.; Zhu, S., DeepGraphGO: graph neural network for large-scale, multispecies protein function prediction. *Bioinformatics* **2021,** *37* (Supplement_1), i262-i271.
133. Kulmanov, M.; Hoehndorf, R., DeepGOZero: improving protein function prediction from sequence and zero-shot learning based on ontology axioms. *Bioinformatics* **2022,** *38* (Supplement_1), i238-i245.
134. Huang, K.; Chandak, P.; Wang, Q.; Havaldar, S.; Vaid, A.; Leskovec, J.; Nadkarni, G.; Glicksberg, B. S.; Gehlenborg, N.; Zitnik, M., Zero-shot drug repurposing with geometric deep learning and clinician centered design. medRxiv: 2023.
135. Gema, A. P.; Grabarczyk, D.; De Wulf, W.; Borole, P.; Alfaro, J. A.; Minervini, P.; Vergari, A.; Rajan, A., Knowledge Graph Embeddings in the Biomedical Domain: Are They Useful? A Look at Link Prediction, Rule Learning, and Downstream Polypharmacy Tasks. arXiv: 2023.
136. Chen, J.; Dong, H.; Hastings, J.; Jiménez-Ruiz, E.; López, V.; Monnin, P.; Pesquita, C.; Škoda, P.; Tamma, V., Knowledge Graphs for the Life Sciences: Recent Developments, Challenges and Opportunities. *TGDK* **2023**.
137. Nicholson, D. N.; Greene, C. S., Constructing knowledge graphs and their biomedical applications. *Computational and Structural Biotechnology Journal* **2020,** *18*, 1414-1428.
138. Li, M. M.; Huang, K.; Zitnik, M., Graph representation learning in biomedicine and healthcare. *Nat. Biomed. Eng* **2022,** *6* (12), 1353-1369.
139. Han, X.; Cao, S.; Lv, X.; Lin, Y.; Liu, Z.; Sun, M.; Li, J. In *OpenKE: An Open Toolkit for Knowledge Embedding*, 2018/11//; Blanco, E.; Lu, W., Eds. Association for Computational Linguistics: 2018; pp 139-144.
140. Biswas, R.; Kaffee, L.-A.; Cochez, M.; Dumbrava, S.; Jendal, T.; Lissandrini, M.; Lopez, V.; Loza Mencia, E.; Paulheim, H.; Sack, H.; Vakaj, E.; De Melo, G., Knowledge Graph Embeddings: Open Challenges and Opportunities. *TGDK* **2023**.
141. Gaudelet, T.; Day, B.; Jamasb, A. R.; Soman, J.; Regep, C.; Liu, G.; Hayter, J. B. R.; Vickers, R.; Roberts, C.; Tang, J.; Roblin, D.; Blundell, T. L.; Bronstein, M. M.; Taylor-King, J. P., Utilizing graph machine learning within drug discovery and development. *Briefings in Bioinformatics* **2021,** *22* (6), bbab159.
142. Morselli Gysi, D.; do Valle, Í.; Zitnik, M.; Ameli, A.; Gan, X.; Varol, O.; Ghiassian, S. D.; Patten, J. J.; Davey, R. A.; Loscalzo, J.; Barabási, A.-L., Network medicine framework for identifying drug-repurposing opportunities for COVID-19. *Proceedings of the National Academy of Sciences of the United States of America* **2021,** *118* (19), e2025581118.
143. Zitnik, M.; Agrawal, M.; Leskovec, J., Modeling polypharmacy side effects with graph convolutional networks. *Bioinformatics* **2018,** *34* (13), i457-i466.
144. Chandak, P.; Huang, K.; Zitnik, M., Building a knowledge graph to enable precision medicine. *Scientific Data* **2023,** *10* (1), 67.

145. Lin, Z.; Akin, H.; Rao, R.; Hie, B.; Zhu, Z.; Lu, W.; Costa, A. d. S.; Fazel-Zarandi, M.; Sercu, T.; Candido, S.; Rives, A., Language models of protein sequences at the scale of evolution enable accurate structure prediction. bioRxiv: 2022.
146. Wang, Z.; Combs, S. A.; Brand, R.; Calvo, M. R.; Xu, P.; Price, G.; Golovach, N.; Salawu, E. O.; Wise, C. J.; Ponnapalli, S. P.; Clark, P. M., LM-GVP: an extensible sequence and structure informed deep learning framework for protein property prediction. *Scientific Reports* **2022,** *12* (1), 6832.
147. Wang, Y.; You, Z.-H.; Yang, S.; Li, X.; Jiang, T.-H.; Zhou, X., A High Efficient Biological Language Model for Predicting Protein–Protein Interactions. *Cells* **2019,** *8* (2), 122.
148. Sledzieski, S.; Singh, R.; Cowen, L.; Berger, B., Adapting protein language models for rapid DTI prediction. bioRxiv: 2022.
149. Kalakoti, Y.; Yadav, S.; Sundar, D., TransDTI: Transformer-Based Language Models for Estimating DTIs and Building a Drug Recommendation Workflow. *ACS Omega* **2022,** *7* (3), 2706-2717.
150. Zhou, G.; Gao, Z.; Ding, Q.; Zheng, H.; Xu, H.; Wei, Z.; Zhang, L.; Ke, G., Uni-Mol: A Universal 3D Molecular Representation Learning Framework. ChemRxiv: 2022.
151. Jiang, D.; Wu, Z.; Hsieh, C.-Y.; Chen, G.; Liao, B.; Wang, Z.; Shen, C.; Cao, D.; Wu, J.; Hou, T., Could graph neural networks learn better molecular representation for drug discovery? A comparison study of descriptor-based and graph-based models. *Journal of cheminformatics* **2021,** *13*, 1-23.
152. Poggio, T.; Mhaskar, H.; Rosasco, L.; Miranda, B.; Liao, Q., Why and when can deep-but not shallow-networks avoid the curse of dimensionality: a review. *International Journal of Automation and Computing* **2017,** *14* (5), 503-519.
153. Zhang, S.; Zhou, J.; Hu, H.; Gong, H.; Chen, L.; Cheng, C.; Zeng, J., A deep learning framework for modeling structural features of RNA-binding protein targets. *Nucleic Acids Res* **2016,** *44* (4), e32.
154. Liu, S.; Wang, H.; Liu, W.; Lasenby, J.; Guo, H.; Tang, J., Pre-training molecular graph representation with 3D geometry. *arXiv [cs.LG]* **2021**.
155. Liu, S.; Guo, H.; Tang, J., Molecular geometry pretraining with SE(3)-invariant denoising distance matching. *arXiv [cs.LG]* **2022**.
156. Rifaioglu, A. S.; Cetin Atalay, R.; Cansen Kahraman, D.; Doğan, T.; Martin, M.; Atalay, V., MDeePred: novel multi-channel protein featurization for deep learning-based binding affinity prediction in drug discovery. *Bioinformatics* **2021,** *37* (5), 693-704.
157. You, Y.; Shen, Y., Cross-modality and self-supervised protein embedding for compound–protein affinity and contact prediction. *Bioinformatics* **2022,** *38* (Supplement_2), ii68-ii74.
158. Tan, H.; Bansal, M., LXMERT: Learning Cross-Modality Encoder Representations from Transformers. *arXiv [cs.CL]* **2019**.
159. Jones, D.; Kim, H.; Zhang, X.; Zemla, A.; Stevenson, G.; Bennett, W. F. D.; Kirshner, D.; Wong, S. E.; Lightstone, F. C.; Allen, J. E., Improved Protein–Ligand Binding Affinity Prediction with Structure-Based Deep Fusion Inference. *Journal of Chemical Information and Modeling* **2021,** *61* (4), 1583-1592.
160. Lam, H. T.; Sbodio, M. L.; Galindo, M. M.; Zayats, M.; Fernández-Díaz, R.; Valls, V.; Picco, G.; Ramis, C. B.; López, V. In *Otter-Knowledge: multimodal knowledge-enhanced representation learning for drug discovery*, AAAI 2024, 2023/10/19/; AAAI Publication: 2023.
161. Pezeshkpour, P.; Chen, L.; Singh, S. In *Embedding Multimodal Relational Data for Knowledge Base Completion*, EMNLP 2018, 2018/10//; Riloff, E.; Chiang, D.; Hockenmaier, J.; Tsujii, J. i., Eds. Association for Computational Linguistics: 2018; pp 3208-3218.
162. Wang, Z.; Wang, Z.; Srinivasan, B.; Ioannidis, V. N.; Rangwala, H.; Anubhai, R., BioBridge: Bridging Biomedical Foundation Models via Knowledge Graphs. arXiv: 2023.
163. Lin, Z.; Akin, H.; Rao, R.; Hie, B.; Zhu, Z.; Lu, W.; Smetanin, N.; Verkuil, R.; Kabeli, O.; Shmueli, Y.; dos Santos Costa, A.; Fazel-Zarandi, M.; Sercu, T.; Candido, S.; Rives, A., Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science* **2023,** *379* (6637), 1123-1130.

164. Gu, Y.; Tinn, R.; Cheng, H.; Lucas, M.; Usuyama, N.; Liu, X.; Naumann, T.; Gao, J.; Poon, H., Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing. *HEALTH* **2021,** *3* (1), 2:1-2:23.
165. Schlichtkrull, M.; Kipf, T. N.; Bloem, P.; van den Berg, R.; Titov, I.; Welling, M. In *Modeling Relational Data with Graph Convolutional Networks*, 2018; Gangemi, A.; Navigli, R.; Vidal, M.-E.; Hitzler, P.; Troncy, R.; Hollink, L.; Tordai, A.; Alam, M., Eds. Springer International Publishing: 2018; pp 593-607.
166. Huang, K.; Fu, T.; Gao, W.; Zhao, Y.; Roohani, Y.; Leskovec, J.; Coley, C.; Xiao, C.; Sun, J.; Zitnik, M., Therapeutics Data Commons: Machine Learning Datasets and Tasks for Drug Discovery and Development. *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks* **2021,** *1*.
167. Staar, P. W. J.; Dolfi, M.; Auer, C., Corpus processing service: A Knowledge Graph platform to perform deep data exploration on corpora. *Applied AI Letters* **2020,** *1* (2), e20.
168. Cramer, R. D.; Patterson, D. E.; Bunce, J. D., Comparative molecular field analysis (CoMFA). 1. Effect of shape on binding of steroids to carrier proteins. *J Am Chem Soc* **1988,** *110* (18), 5959-67.
169. Klebe, G.; Abraham, U.; Mietzner, T., Molecular similarity indices in a comparative analysis (CoMSIA) of drug molecules to correlate and predict their biological activity. *J Med Chem* **1994,** *37* (24), 4130-46.
170. Todeschini, R.; Moro, G.; Boggia, R.; Bonati, L.; Cosentino, U.; Lasagni, M.; Pitea, D., Modeling and prediction of molecular properties. Theory of grid-weighted holistic invariant molecular (G-WHIM) descriptors. *Chemometrics and Intelligent Laboratory Systems* **1997,** *36* (1), 65-73.
171. Cruciani, G.; Crivori, P.; Carrupt, P. A.; Testa, B., Molecular fields in quantitative structure–permeation relationships: the VolSurf approach. *Journal of Molecular Structure: THEOCHEM* **2000,** *503* (1), 17-30.
172. Tortorella, S.; Carosati, E.; Sorbi, G.; Bocci, G.; Cross, S.; Cruciani, G.; Storchi, L., Combining machine learning and quantum mechanics yields more chemically aware molecular descriptors for medicinal chemistry applications. *J Comput Chem* **2021,** *42* (29), 2068-2078.
173. Nha Tran, T. T.; Thuan Tran, T. D.; Thuy Bui, T. T., Integration of machine learning in 3D-QSAR CoMSIA models for the identification of lipid antioxidant peptides. *RSC Adv* **2023,** *13* (48), 33707-33720.
174. Hopfinger, A. J.; Wang, S.; Tokarski, J. S.; Jin, B.; Albuquerque, M.; Madhav, P. J.; Duraiswami, C., Construction of 3D-QSAR Models Using the 4D-QSAR Analysis Formalism. *Journal of the American Chemical Society* **1997,** *119* (43), 10509-10524.
175. Smith, J. S.; Isayev, O.; Roitberg, A. E., ANI-1, A data set of 20 million calculated off-equilibrium conformations for organic molecules. *Scientific Data* **2017,** *4* (1), 170193.
176. Groom, C. R.; Bruno, I. J.; Lightfoot, M. P.; Ward, S. C., The Cambridge Structural Database. *Acta Crystallographica Section B* **2016,** *72* (2), 171-179.
177. Berman, H. M.; Battistuz, T.; Bhat, T. N.; Bluhm, W. F.; Bourne, P. E.; Burkhardt, K.; Feng, Z.; Gilliland, G. L.; Iype, L.; Jain, S.; Fagan, P.; Marvin, J.; Padilla, D.; Ravichandran, V.; Schneider, B.; Thanki, N.; Weissig, H.; Westbrook, J. D.; Zardecki, C., The Protein Data Bank. *Acta Crystallogr. D Biol. Crystallogr.* **2002,** *58* (6), 899-907.
178. Rodriguez-Espigares, I.; Torrens-Fontanals, M.; Tiemann, J. K. S.; Aranda-Garcia, D.; Ramirez-Anguita, J. M.; Stepniewski, T. M.; Worp, N.; Varela-Rial, A.; Morales-Pastor, A.; Medel-Lacruz, B.; Pandy-Szekeres, G.; Mayol, E.; Giorgino, T.; Carlsson, J.; Deupi, X.; Filipek, S.; Filizola, M.; Gomez-Tamayo, J. C.; Gonzalez, A.; Gutierrez-de-Teran, H.; Jimenez-Roses, M.; Jespers, W.; Kapla, J.; Khelashvili, G.; Kolb, P.; Latek, D.; Marti-Solano, M.; Matricon, P.; Matsoukas, M. T.; Miszta, P.; Olivella, M.; Perez-Benito, L.; Provasi, D.; Rios, S.; I, R. T.; Sallander, J.; Sztyler, A.; Vasile, S.; Weinstein, H.; Zachariae, U.; Hildebrand, P. W.; De Fabritiis, G.; Sanz, F.; Gloriam, D. E.; Cordomi, A.; Guixa-Gonzalez, R.; Selent, J., GPCRmd uncovers the dynamics of the 3D-GPCRome. *Nat Methods* **2020,** *17* (8), 777-787.
179. MolSSI, B. COVID-19 Molecular Structure and Therapeutics Hub. https://covid.bioexcel.eu/.

180. Pan, S.; Luo, L.; Wang, Y.; Chen, C.; Wang, J.; Wu, X., Unifying Large Language Models and Knowledge Graphs: A Roadmap. arXiv: 2023.
181. Wang, X.; Chen, W. In *Knowledge Graph Data Management: Models, Methods, and Systems*, 2020; U, L. H.; Yang, J.; Cai, Y.; Karlapalem, K.; Liu, A.; Huang, X., Eds. Springer: 2020; pp 3-12.
182. Timón-Reina, S.; Rincón, M.; Martínez-Tomás, R., An overview of graph databases and their applications in the biomedical domain. *Database* **2021,** *2021*, baab026.
183. Fey, M.; Lenssen, J. E.; Weichert, F.; Leskovec, J. In *GNNAutoScale: Scalable and Expressive Graph Neural Networks via Historical Embeddings*, International Conference on Machine Learning, 2021/07/01/; PMLR: 2021; pp 3294-3304.
184. Mysinger, M. M.; Carchia, M.; Irwin, J. J.; Shoichet, B. K., Directory of useful decoys, enhanced (DUD-E): better ligands and decoys for better benchmarking. *Journal of medicinal chemistry* **2012,** *55* (14), 6582-6594.
185. Chen, L.; Cruz, A.; Ramsey, S.; Dickson, C. J.; Duca, J. S.; Hornak, V.; Koes, D. R.; Kurtzman, T., Hidden bias in the DUD-E dataset leads to misleading performance of deep learning in structure-based virtual screening. *PloS one* **2019,** *14* (8), e0220113.
186. Klarner, L.; Reutlinger, M.; Schindler, T.; Deane, C.; Morris, G. In *Bias in the Benchmark: Systematic experimental errors in bioactivity databases confound multi-task and meta-learning algorithms*, ICML 2022 2nd AI for Science Workshop, 2022.
187. Moult, J.; Fidelis, K.; Kryshtafovych, A.; Schwede, T.; Tramontano, A., Critical assessment of methods of protein structure prediction (CASP)—round x. *Proteins: Structure, Function, and Bioinformatics* **2014,** *82*, 1-6.
188. Janin, J.; Henrick, K.; Moult, J.; Eyck, L. T.; Sternberg, M. J.; Vajda, S.; Vakser, I.; Wodak, S. J., CAPRI: a critical assessment of predicted interactions. *Proteins: Structure, Function, and Bioinformatics* **2003,** *52* (1), 2-9.
189. Zhou, N.; Jiang, Y.; Bergquist, T. R.; Lee, A. J.; Kacsoh, B. Z.; Crocker, A. W.; Lewis, K. A.; Georghiou, G.; Nguyen, H. N.; Hamid, M. N., The CAFA challenge reports improved protein function prediction and new functional annotations for hundreds of genes through experimental screens. *Genome biology* **2019,** *20* (1), 1-23.
190. Parks, C. D.; Gaieb, Z.; Chiu, M.; Yang, H.; Shao, C.; Walters, W. P.; Jansen, J. M.; McGaughey, G.; Lewis, R. A.; Bembenek, S. D., D3R grand challenge 4: blind prediction of protein–ligand poses, affinity rankings, and relative binding free energies. *Journal of computer-aided molecular design* **2020,** *34*, 99-119.
191. Karim, M. R.; Islam, T.; Beyan, O.; Lange, C.; Cochez, M.; Rebholz-Schuhmann, D.; Decker, S., Explainable AI for Bioinformatics: Methods, Tools, and Applications. arXiv: 2022.